

Do Substance Rules Have Substance? Firm Responses to Anti-Tax Avoidance Provisions*

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Abstract

How do multinational firms respond to anti-tax avoidance provisions? We focus on substance rules, which have become a central guardrail of modern international tax regimes, and allow firms to avoid the application of anti-avoidance rules if they demonstrate economic activity in the low-tax jurisdictions where they report profits. Using two European regulatory changes to Controlled Foreign Company (CFC) rules, we examine firm responses along two dimensions, where firms locate their subsidiaries and how they adjust economic activity within them. First, using the 2006 Cadbury Schweppes ruling, we show that firms respond to substance-based exemptions by shifting where they locate subsidiaries. Because the exemption lets firms avoid CFC taxation by establishing substance in a low-tax jurisdiction, affected firms increase both the number and share of their subsidiaries there. Second, leveraging the 2016 adoption of the Anti-Tax Avoidance Directive (ATAD), we examine how multinationals meet substance requirements in affiliates they already operate. Firms increase personnel costs in low-tax affiliates without adding employees or expanding investment, while related subsidiaries in higher-tax countries experience offsetting declines in personnel costs. Overall, our findings suggest that firms use substance-based exemptions to expand their presence in low-tax jurisdictions, and satisfy the requirements by reallocating activity within their network rather than by creating new activity where they report profits.

Keywords: Substance, Economic activity, Anti-tax avoidance, Multinational firms, Corporate taxation

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1 Introduction

Multinational firms have considerable latitude in deciding where to report the profits they earn. Prior literature shows that many firms shift pre-tax income toward low-tax jurisdictions to minimize their tax payments, a strategy made possible by countries competing for mobile capital through low corporate tax rates ([Dyreng and Hanlon, 2021](#)). This behavior is large in scale and well documented, with estimates suggesting that multinationals shift roughly 35 to 40 percent of their global profits away from the countries where their real economic activity takes place ([Clausing, 2020](#); [Tørsløv et al., 2023](#); [Blouin and Robinson, 2025](#)).¹ Such large-scale income shifting erodes government revenue and raises the question of whether firms pay tax where they create economic value. In response, governments worldwide have introduced anti-tax avoidance rules that specifically target income reported in low-tax jurisdictions. In this paper, we study how multinational firms respond to substance rules, provisions that condition these anti-avoidance rules on the presence of real economic activity.

Anti-tax avoidance rules generally impose a tax penalty on income reported in low-tax jurisdictions, regardless of whether that income reflects real business activity or purely tax-motivated income shifting. Controlled Foreign Company (CFC) rules are a common example. They attribute the income of low-taxed foreign subsidiaries to their parent company and subject it to immediate taxation at the parent's higher rate. This approach creates a fundamental tension, because it taxes all low-tax income equally without regard to substance, penalizing genuine foreign investment and legitimate business location decisions alongside income shifting. In the European Union (EU), this tension had direct legal consequences, as the European Court of Justice found in its *Cadbury Schweppes* ruling in 2006 that anti-tax avoidance rules could not be applied indiscriminately to low-tax subsidiaries carrying out genuine economic activity, since doing so would restrict the freedom of establishment guaranteed under EU law ([European Court of Justice, 2006](#)). Policymakers in the EU and elsewhere responded to this by adding substance-based exemptions to anti-tax avoidance regulations, carving out an exception for firms that demonstrate real economic activity.²

Substance rules have since become a central guardrail of modern anti-tax avoidance policy. Across a growing number of regimes, including the U.S. Global Intangible Low-Taxed Income (GILTI) rules, CFC regulations, and the OECD's Global Minimum

¹Estimates of the magnitude of global profit shifting vary depending on the data and methodology used. Firm-level approaches tend to yield lower estimates than macro-level approaches, and the exact magnitude remains subject to debate, though studies agree that the phenomenon is economically significant ([Clausing, 2020](#); [Blouin and Robinson, 2025](#)).

²International policymakers have since echoed similar reasoning. In its Base Erosion and Profit Shifting (BEPS) action plan, the OECD explained that a "re-alignment of taxation and relevant substance" was needed to restore the intended effects of international tax standards ([OECD, 2013](#)). This reasoning is also underlying the subsequent extension of CFC rules in the EU with substance-based exemptions under the Anti-Tax Avoidance Directive in 2016.

Tax, these provisions exempt firms from anti-avoidance taxation if they demonstrate sufficient economic activity, such as employees and physical assets, in the jurisdiction where they report profits. In principle, this guardrail ties taxation to real economic value creation, discouraging purely paper-based income shifting while leaving genuine business activity unaffected. Yet little is known about whether firms respond to substance rules as policymakers intend, or instead adapt in ways that undermine this goal. We study these firm responses along two dimensions. Specifically, we analyze (1) whether firms shift the location of their subsidiaries toward low-tax jurisdictions to retain the tax advantages that substance-based exemptions newly allow them to retain, and (2) how they adjust economic activity within existing affiliates once these become subject to substance requirements.

Studying firm responses to substance rules is important because these provisions fundamentally change the incentives for income shifting. Traditional anti-avoidance rules impose a tax penalty on income reported in low-tax jurisdictions as a way to reduce income shifting. Prior literature documents that firms respond to these rules by reducing investment, closing subsidiaries or relocating economic activity to higher-tax jurisdictions (see e.g., [Voget, 2011](#); [Clifford, 2019](#); [Bilicka et al., 2022](#)). Substance-based exemptions alter this dynamic by conditioning the tax penalty on the absence of economic activity. This conditionality creates new and competing incentives. On the one hand, firms may respond as intended, by locating real economic activity in the low-tax jurisdictions where they report income. On the other hand, firms may instead use substance rules to legitimize income shifting, either by increasing their presence in low-tax jurisdictions or by reallocating existing activity from higher-tax to lower-tax affiliates to escape the anti-tax avoidance rule. Which response dominates is ultimately an empirical question, and the answer carries direct implications for the effectiveness of these provisions. If substance rules primarily attract subsidiaries into low-tax jurisdictions without generating new economic value, they may facilitate income shifting rather than restrict it. Moreover, if firms meet substance requirements by reallocating activity rather than expanding it, countries implementing anti-avoidance rules may face not only lower tax revenues but also declines in local investment and employment. Thus, empirical evidence is critical for informing the effectiveness of these types of income shifting regulations.

We examine firm responses to substance-based exemptions in the context of CFC rules in the EU. CFC rules are designed to discourage income shifting by attributing the passive income of low-tax foreign subsidiaries to their higher-tax parent companies and subjecting that income to immediate taxation at the parent level.³ These rules provide an ideal laboratory to study firm responses to substance rules for three reasons. First, CFC rules apply across a broad set of countries, within and beyond the EU, making

³Passive income is generally characterized either as highly mobile income streams, including dividends, interest, and royalties, or more broadly as income arising from non-genuine arrangements.

the findings relevant well outside our specific setting. Second, the notion of substance in CFC regimes is vaguely defined, without precise thresholds for the activity a subsidiary must demonstrate. This lack of clarity makes these rules especially prone to both strategic reporting and real responses. Third, two major regulatory changes in the EU, the Cadbury Schweppes ruling in 2006 and the Anti-Tax Avoidance Directive (ATAD) in 2016, introduced substance-based exemptions at different times and to different groups of firms. Both create quasi-experimental variation we can exploit to best identify the effects of substance-based rules and distinguish between different features of the policy.

We use the Cadbury Schweppes ruling to study our first research question, whether firms shift the location of their subsidiaries toward low-tax jurisdictions following the introduction of substance-based exemptions. The ruling required countries with existing CFC rules to introduce a substance-based exemption. This could make low-tax jurisdictions newly attractive to multinationals, since firms can retain the tax advantages of these jurisdictions by demonstrating substance there. We use the second reform, the ATAD, to examine our second research question, how multinationals adjust economic activity within existing affiliates once these become subject to substance requirements. The directive extended CFC rules with substance requirements to member states that had not previously had CFC rules in place. This confronted existing subsidiaries of multinationals with CFC rules including substance requirements they had not previously faced, which is in contrast to the Cadbury Schweppes ruling which applied only where CFC rules already existed. Thus, we view the two settings as complementary. The Cadbury Schweppes setting reveals where firms locate activity but cannot show whether they build real activity into these subsidiaries, since the subsidiaries we study are established only after the ruling and therefore have no pre-reform baseline against which to measure changes in their activity.⁴ In contrast, the ATAD setting provides exactly this baseline, allowing us to observe how existing subsidiaries adjust from a clean before-and-after comparison. We view both settings together as providing a holistic and complementary view.

We begin with the Cadbury Schweppes setting, examining whether firms shift the location of their subsidiaries toward low-tax jurisdictions. The income shifting literature documents that firms locate subsidiaries in low-tax jurisdictions, known as tax havens, to minimize their tax payments (e.g., [Hines and Rice, 1994](#); [Dyreng and Lindsey, 2009](#); [Dharmapala, 2014](#)). Prior research on anti-tax avoidance regulations, in particular on CFC rules before the implementation of substance-based exemptions, shows that they lead firms to close or relocate existing subsidiaries ([Voget, 2011](#); [Clifford, 2019](#)). The introduction of substance exemptions may reverse this by allowing firms to keep the tax advantages of a low-tax subsidiary as long as it demonstrates sufficient eco-

⁴Nevertheless, we provide motivational descriptive evidence on the economic activity of these newly incorporated subsidiaries. This evidence is purely descriptive and does not establish a causal effect.

conomic activity ([Williams, 2018](#); [De Simone and Olbert, 2022](#); [Drake et al., 2022](#)). Thus, the substance-based exemption could make these jurisdictions attractive again, potentially opening a new channel for income shifting. Whether firms actually respond, however, is not obvious. Locating and substantiating a subsidiary in a low-tax jurisdiction entails administrative, compliance, and operational costs, as well as reputational concerns, and firms will only do so if the tax savings outweigh these costs ([Scholes and Wolfson, 1992](#); [Graham et al., 2014](#); [Gallemore et al., 2014](#)).

To test this prediction, we use financial statement and ownership data from Moody's Orbis database and employ a difference-in-differences (DiD) research design around the Cadbury Schweppes ruling, which added substance-based exemptions to existing CFC legislation. We compare multinationals headquartered in EU countries with pre-existing CFC rules, which were required to introduce substance-based exemptions following the ruling, to multinationals headquartered in EU countries without CFC rules, over the period 2002 to 2008. Because the ruling changed incentives specifically for firms deciding whether to locate subsidiaries in low-tax jurisdictions, we examine how the number and share of low-tax CFC subsidiaries held by treated multinationals change relative to the control group.

We find that treated multinationals increase the number of CFC subsidiaries by 4.6% following the ruling, consistent with substance-based exemptions attracting new subsidiary locations to low-tax jurisdictions. Turning to the composition of multinationals' subsidiary portfolios, we find that the share of CFC subsidiaries rises by roughly 7.5%, with a corresponding decline in the share of non-CFC subsidiaries. In a fractional logit specification, we estimate that the likelihood that a treated multinational locates a subsidiary in a low-tax jurisdiction rather than a higher-tax jurisdiction increases by 10% relative to control multinationals. We interpret this pattern as suggesting that firms do not only expand their presence in low-tax jurisdictions, but also redirect a greater share of their subsidiaries toward them. Event study DiD graphs around the ruling show no differential pre-treatment trends for either outcome, supporting the parallel trends assumption underlying our design. We further confirm the robustness of our results across log, inverse hyperbolic sine (IHS), level, logit, and Poisson pseudo-maximum-likelihood (PPML) specifications, alternative fixed-effects structures, and leave-one-out analyses excluding individual host countries.

In cross-sectional tests, we shed light on the mechanisms behind these results. If firms respond to the substance-based exemption by establishing subsidiaries in low-tax countries, the response should be strongest where the exemption relaxes the tightest constraint. We therefore split the sample by the tax rate threshold that triggers CFC taxation in the parent's home country. Multinationals headquartered in high-threshold countries face the tighter constraint, since a broader set of low-tax jurisdictions falls under their CFC rules, leaving them little room to locate subsidiaries there without

triggering CFC taxation. For these firms, the exemption opens an avenue that was previously unavailable to them, provided they demonstrate sufficient substance. Consistent with this, we find that the increase in the count and share of low-tax subsidiaries is concentrated among multinationals from high-threshold countries. We also find the response concentrated among smaller multinationals and those operating narrower subsidiary networks before the reform, consistent with the exemption opening access to low-tax jurisdictions for firms that previously had few routes to them. Larger and more complex multinationals show no significant response, in line with prior evidence that these firms already have income shifting arrangements available to them (Rego, 2003; Balakrishnan et al., 2019; Francois and Vicard, 2025). Together, these patterns indicate that firms respond to the substance-based exemption where the exemption represents the largest change in their opportunity set.

The Cadbury Schweppes setting does not allow us to observe whether these newly established subsidiaries carry out sufficient economic activity, since they exist only after the ruling and therefore have no pre-reform baseline. As motivational evidence, we nevertheless compare newly established treated subsidiaries to newly established untreated subsidiaries descriptively in the years after incorporation, examining whether they have the substance to qualify for the exemption, since only subsidiaries that satisfy the substance requirement can retain the tax benefits of a low-tax location. We measure substance using the level of fixed assets, employees, and personnel costs. We find that subsidiaries of treated multinationals hold significantly more fixed assets and incur higher personnel costs. While purely descriptive, this pattern is consistent with treated multinationals establishing subsidiaries with sufficient substance to meet the exemption threshold.

The ATAD setting allows us to move beyond this descriptive evidence, since we can follow existing low-tax subsidiaries before and after they become subject to CFC rules with substance requirements. We use this second setting to study whether and how firms adjust economic activity within these subsidiaries. To keep the tax advantages of a low-tax subsidiary while avoiding the CFC penalty, firms must demonstrate sufficient substance in that subsidiary. What constitutes sufficient substance, however, is often defined only in vague terms, such as maintaining "staff, equipment, assets, and premises" (European Union, 2016). This general definition leaves firms with considerable scope in how they respond. One possibility is that firms respond as intended by building up real economic activity, such as expanding investment and employment in the subsidiary (Williams, 2018; De Simone and Olbert, 2022; Drake et al., 2022). Another possibility is that firms respond with reporting-based adjustments rather than real changes (Lester, 2019; Chen et al., 2021; Kelley et al., 2024). For example, firms may satisfy the requirement through changes in workforce composition or reporting arrangements, such as allocating higher-paid personnel to low-tax affiliates, rather than

through real operational expansion ([Bornemann et al., 2023](#); [Chen et al., 2023](#)). Even where firms build up real economic activity in the low-tax subsidiary, this activity need not reflect new value creation, since firms could instead shift investment and employment away from related subsidiaries in higher-tax jurisdictions, satisfying the substance requirement by reallocating existing activity rather than expanding it. Whether firms respond by creating real economic activity or by adjusting along strategic reporting margins, and whether any response is new or reallocated, speaks directly to whether substance rules achieve their intended alignment of taxation with real value creation.

In the ATAD setting, we compare subsidiaries of multinationals that become subject to CFC rules with substance-based exemptions for the first time to subsidiaries of multinationals already subject to such rules, over the period 2012 to 2019. As in the Cadbury Schweppes setting, we use financial statement and ownership data from Moody's Orbis database and employ a DiD research design. This lets us examine how newly affected subsidiaries adjust their reported investment and employment, which we measure using fixed assets, the number of employees, and personnel costs. To assess whether any increases in investment and employment reflect genuine expansion or a reallocation of activity within the multinational group, we additionally examine subsidiaries of the same groups located in higher-tax countries, which fall outside the scope of CFC rules.

We find no significant effect on fixed assets or on the number of employees of CFC subsidiaries, indicating that firms do not respond to substance requirements by expanding investment or hiring additional staff. Instead, the response is concentrated in personnel costs, which rise by 3.4% (log specification; 9.4% using PPML). In absolute terms, the average treated subsidiary adds roughly EUR 97,000 in personnel expenses (EUR 269,000 under PPML) without adding a single employee. The dynamic estimates reinforce this pattern. The number of employees increases only in the first year after treatment and reverts thereafter, whereas the increase in personnel costs persists throughout the post-treatment period. This stronger and more persistent response in personnel costs than in headcount indicates that firms do not simply add workers, but shift toward higher-paid personnel or reallocate reported personnel to low-tax affiliates, consistent with a reporting-based response rather than real operational expansion. Our event studies support the parallel trends assumption, showing no differential pre-treatment trends for either outcome. We further confirm the robustness of our results across log, IHS, and PPML specifications, and alternative fixed-effects structures.

To assess whether these increases reflect real expansion or a reallocation of activity within the group, we examine subsidiaries of the same multinationals located in higher-tax countries. We find offsetting declines in personnel costs among these non-CFC subsidiaries, indicating that firms shift labor expenses from higher-tax to low-

tax affiliates rather than adding them. This mirrors the pattern we document in the low-tax affiliates, where personnel costs rise without a corresponding change in headcount, and points toward adjustments in reported personnel rather than real operational change, though our data do not allow us to fully disentangle a shift in real activity from a shift in its reporting.

We further examine which subsidiaries drive these responses. If firms adjust economic activity specifically to satisfy substance requirements, the response should be strongest among subsidiaries with the most to gain from qualifying for the exemption. First, the benefits of qualifying should be greater for subsidiaries facing a larger tax differential between the parent and host country, since more tax is at stake. Consistent with this, the increase in personnel costs is concentrated among subsidiaries with larger tax rate differentials relative to their parent. Second, subsidiaries with a higher share of passive income have more income exposed to CFC taxation, and therefore face a higher penalty if they fail the substance test. Confirming this, the effect is more pronounced for subsidiaries holding a higher share of passive income. Third, subsidiaries with lower pre-existing levels of activity must adjust the most to satisfy the substance requirement, and we find that these subsidiaries respond more strongly than those that already maintain substantial activity. For the first two splits, we additionally find positive effects on the number of employees, indicating that where the incentive to qualify is strongest, firms add staff alongside personnel costs rather than relying on costs alone. We find no heterogeneity in fixed assets across any of the three splits, reinforcing that firms respond through labor rather than capital. Overall, firms adjust where they have the most income at stake and the most to adjust to meet the requirement, indicating that they act specifically to protect income that would otherwise fall under CFC taxation.

Our study makes two contributions. First, we contribute to the literature on the real and reporting effects of anti-tax avoidance regulation, responding to recent calls for research on how firms adapt to these rules ([Jacob, 2022](#); [Lester and Olbert, 2025](#)). Prior work has examined firm responses to a range of anti-tax avoidance measures, including interest deduction limitations, CFC rules, general anti-abuse provisions, and disclosure requirements, with most attention devoted to real effects ([Clifford, 2019](#); [Joshi, 2020](#); [Bilicka et al., 2022](#); [De Simone and Olbert, 2022](#); [Cowx and Kerr, 2024](#)) and comparatively little to reporting effects. We depart from this literature in two ways. We focus on substance-based exemptions, a design feature that has become central to many anti-avoidance regimes but remains largely unstudied in its own right, and we add a reporting margin to this literature, following recent tax accounting work ([Lester, 2019](#); [Kelley et al., 2024](#)). Prior evidence shows that CFC rules without substance-based exemptions deter multinationals from locating subsidiaries in low-tax jurisdictions ([Vogel, 2011](#); [Clifford, 2019](#)) and reduce investment and employment where they already

operate (Egger and Wamser, 2015; Albertus, 2025).⁵ We find a different pattern once an exemption is introduced. Multinationals increase both the number and share of subsidiaries they locate in low-tax jurisdictions, and personnel costs rise while employment and investment show no change. The increase in personnel costs without a corresponding increase in headcount, together with offsetting declines in higher-tax affiliates, indicates, though it does not fully establish, that firms meet substance requirements in part by reallocating and reclassifying existing personnel rather than through real operational change.

Second, we contribute to the literature on how firms respond to tax provisions that condition benefits on economic activity. One strand of literature shows that firms substantiate income shifting with real activity, building up employment and investment in low-tax jurisdictions to justify the income reported there (Williams, 2018; De Simone and Olbert, 2022; Drake et al., 2022). A second strand examines preferential tax regimes that explicitly condition tax benefits on economic activity, such as intellectual property boxes and employment requirements in tax zones, and finds that firms shift workforce composition toward higher-paid, specialized employees rather than expanding headcount broadly (Chen et al., 2023; Bornemann et al., 2023; Cabral et al., 2025). We bridge these two strands by studying both whether and how firms substantiate their presence in low-tax jurisdictions. We find that multinationals build up their presence in low-tax jurisdictions once substance exemptions are introduced, and that this substantiation is concentrated in personnel costs rather than headcount.

Our findings speak to tax authorities and policymakers designing anti-tax avoidance regimes. While our analysis focuses on CFC rules, the insights extend to any regime that conditions tax treatment on economic activity, including the GILTI rules and the Global Minimum Tax. Three implications arise. First, substance-based exemptions can undo the deterrent effect of traditional anti-avoidance rules, drawing subsidiary activity back toward low-tax jurisdictions. Second, because firms appear to meet substance requirements by raising personnel costs alone, without any change in headcount or investment, requirements defined only in terms of maintaining staff may be satisfiable without real operational change. Policymakers aiming to tie taxation more closely to real activity may need to define substance more precisely or pair it with additional conditions that are harder to satisfy through reporting adjustments alone.

⁵A broader literature studies the effects of CFC rules on income shifting, measured through foreign pre-tax earnings, or financial assets, as well as on foreign direct investment and M&A activity (von Hagen and Harendt, 2017; Prettl, 2018; Prettl and von Hagen, 2023). These studies examine settings prior to the introduction of substance-based exemptions. Three studies leverage the Cadbury Schweppes setting specifically, but each addresses a different outcome. Ruf and Weichenrieder (2013) find that German multinationals increased passive assets in low-tax subsidiaries following the ruling; Schenkelberg (2020) find a corresponding increase in income shifting within European subsidiaries; and Davies et al. (2026) study the ruling's effect on domestic employment in the multinational's home country. None of these studies examine whether the ruling changed where multinationals locate subsidiaries in the first place.

Third, because the firms meet these requirements by drawing on higher-tax affiliates in the group, these higher-tax affiliate countries adopting substance-based exemptions may face not only lower tax revenues but also reductions in taxable compensation.

2 Background and Predictions

2.1 Substance Rules in CFC Provisions

Substance rules serve as a policy guardrail in a wide range of anti-tax avoidance regimes, including the U.S. GILTI rules, CFC provisions, and the OECD's Global Minimum Tax. Broadly, these rules exempt firms from anti-tax avoidance provisions when sufficient economic activity can be demonstrated. Their purpose is to target arrangements primarily designed for income shifting rather than for genuine economic reasons. In practice, substance is defined in terms of investment and employment, as under GILTI and the Global Minimum Tax, or more loosely as the presence of "people, premises, assets, and risks" in the context of CFC provisions (OECD, 2015a). Importantly, substance rules differ in their design. While GILTI and the Global Minimum Tax apply percentage-based carve-outs tied to tangible assets and payroll, providing clear and verifiable thresholds, CFC regimes generally exempt an entity from the application of the rules altogether if sufficient substance can be demonstrated. The absence of precise thresholds in CFC regimes leaves substantial discretion in determining what constitutes sufficient substance, allowing firms latitude in how they respond.

In this paper, we focus on substance rules within CFC provisions in the EU. CFC rules aim to deter multinationals from shifting income from high-tax to low-tax countries by immediately taxing the profit a foreign subsidiary's profit at the parent company's level, provided two conditions are fulfilled. First, the subsidiary must be controlled by the parent company, typically defined as direct or indirect majority ownership of voting rights, capital, or profit entitlements. This control criterion ensures that CFC rules apply only to entities whose decisions can be influenced by the parent and that can therefore be used for income shifting purposes. Second, the subsidiary must be considered low-tax. Low taxation is defined either relative to the parent country's domestic tax level, as an explicit statutory CIT rate threshold, or through blacklists classifying countries as low-tax for CFC purposes.

When both conditions are satisfied, the subsidiary is classified as a CFC, and its passive income is attributed to the parent company. EU countries differ in their definitions of passive income, with some targeting specific categories such as dividends, interest, or royalties, while others apply broader concepts based on non-genuine arrangements. Upon attribution, the passive income of the low-tax subsidiary is taxed at the parent company's higher CIT rate, with foreign taxes paid being credited. Thus, economically, CFC rules impose a tax penalty on income shifting equal to the tax rate differential between the parent company's home country and the subsidiary's host country.

Substance-based exemptions became a mandatory feature of EU CFC regimes following the 2006 Cadbury Schweppes decision of the European Court of Justice ([European Court of Justice, 2006](#)). In this ruling, the Court held that, within the European Economic Area (EEA), CFC rules may only be applied to wholly artificial arrangements, such as letter-box companies ([Bräutigam et al., 2017](#)). Subsidiaries engaged in genuine economic activity therefore fall outside the scope of CFC taxation. As a result, the eleven EU countries with CFC rules in force at the time were required to amend their legislation to incorporate substance-based exemptions for EEA subsidiaries, rendering CFC rules inapplicable to low-tax EEA subsidiaries that conduct real economic activity.⁶ Because the ruling introduced substance-based exemptions for multinationals already operating under CFC rules, we use it to study how firms adjust the location of their subsidiaries in response.

The second major introduction of substance rules occurred with the adoption of the ATAD in 2016, which required the 15 EU member states without existing CFC regimes to implement CFC rules with substance-based exemptions by the end of 2018 ([European Union, 2016](#)).⁷ The directive grants EU countries discretion to choose between two alternative CFC models. One model explicitly exempts EEA subsidiaries that carry out “substantive economic activities” (Article 7 (2) (a) ATAD), while the other incorporates the substance rule implicitly by limiting the application of CFC rules to income arising from “non-genuine arrangements established primarily for the purpose of securing a tax advantage” (Article 7 (2) (b) ATAD). Because the ATAD newly subjected previously unaffected multinationals to substance requirements for subsidiaries they already operated, we use it to study how firms adjust economic activity within existing subsidiaries.

Both the Cadbury Schweppes framework and the ATAD define substance in terms of staff, equipment, assets, and premises, with particular emphasis on qualified personnel who perform meaningful functions and can exercise control over the entity’s activities. However, these criteria lack precise thresholds. The provisions do not specify the number of employees required, the share of time personnel must dedicate to the entity, or whether and how much physical presence is necessary.⁸ Because the requirement specifies neither the amount nor the type of activity needed in a low-tax entity, firms retain discretion over which margins they adjust to demonstrate substance.

⁶Appendix Table A-2 provides an overview of the countries that had CFC rules in force in 2006 and were therefore affected by the Cadbury Schweppes ruling.

⁷Table A-2 in the Appendix shows the countries that implemented CFC rules with substance-based exemptions for the first time.

⁸The ambiguity surrounding substance requirements has given rise to tax consultancies offering solutions designed to meet these requirements. For example, Confidus Solutions advertises “virtual office services” as “an elegant and cost-effective solution” for meeting staff and premises requirements (see <https://www.confiduss.com/en/services/incorporation/location/tax-haven/>). Similarly, other providers offer “management packages” in low-tax jurisdictions (see <https://pmg-offshore-company.com/offshore-freezone-companies>).

2.2 Firm Responses to Substance Rules

Income shifting literature documents that firms shift income from high-tax to low-tax jurisdictions to minimize their tax payments ([Hines and Rice, 1994](#); [Dyreng and Lindsey, 2009](#); [Dharmapala, 2014](#)). Concern for policymakers ([OECD, 2013](#)) implemented anti-tax avoidance regulations, prior research on these shows that firms closing or relocating subsidiaries in low-tax countries and reducing overall investment in affected countries ([Bilicka et al., 2022](#); [Li and Ma, 2022](#); [Fich et al., 2026](#)). CFC rules without substance-based exemptions operate through the same channel. By imposing additional taxation on income earned in low-tax countries, they significantly raise the effective tax rate on such income and thus reduce the incentive to locate and invest in affiliates there. This pattern has been documented in prior work on CFC regimes ([Voget, 2011](#); [Egger and Wamser, 2015](#); [Clifford, 2019](#)).

Substance-based exemptions modify this incentive structure by conditioning the application of the anti-tax avoidance rule on the absence of economic activity. Rather than facing an unconditional tax penalty, firms can preserve the tax advantages of low-tax subsidiaries if they meet the relevant substance criteria. In principle, this design should discourage purely paper-based income shifting and encourage firms to align reported income with economic activity. However, substance-based exemptions may also open new ways for firms to avoid CFC taxation. While prior evidence suggests that income shifting increased following the mandatory integration of substance-based exemptions into CFC rules ([Ruf and Weichenrieder, 2013](#); [Schenkelberg, 2020](#)), these studies do not examine how firms achieve this outcome. It therefore remains an open and policy-relevant empirical question whether substance rules effectively align taxation with real economic activity.⁹

We structure our analysis around two ways in which firms may respond to substance rules. First, we study whether substance-based exemptions change where firms locate their subsidiaries, examining both the number of subsidiaries multinationals hold in low-tax jurisdictions and the share these represent of their total subsidiaries. Second, we study how firms adjust economic activity within existing low-tax subsidiaries once these become subject to CFC rules with substance-based exemptions, looking at whether firms build up investment and employment to meet the substance requirements. For each part, we further examine the mechanisms that drive the responses. Together, these two parts allow us to assess the overall effectiveness of substance-based exemptions in anti-tax avoidance policies.

Our first hypothesis examines whether the introduction of substance rules increases the attractiveness of low-tax countries as locations for subsidiaries. Prior research shows that CFC rules without substance-based exemptions reduce the incentive to locate sub-

⁹Reflecting these concerns, policymakers and tax authorities raised early concerns that substance rules could create a safe harbor for tax planning within the EU following their initial introduction ([OECD, 2015b](#)).

subsidiaries in low-tax countries (Voget, 2011; Clifford, 2019). If firms can avoid CFC taxation by demonstrating sufficient economic activity, low-tax countries may again become attractive locations. In particular, firms may find it profitable to establish subsidiaries in low-tax jurisdictions if doing so allows them to shift income while meeting substance requirements. Accordingly, we formulate our first hypothesis as follows:

Hypothesis 1: Following the introduction of substance-based exemptions, multinationals subject to CFC rules increase their subsidiary activity in low-tax countries.

We test this hypothesis using two outcomes that capture different aspects of the same shift. The count of low-tax subsidiaries captures whether multinationals expand their presence in low-tax jurisdictions in absolute terms, while the share captures whether they redirect a greater proportion of their subsidiary network toward these jurisdictions relative to their total activity. The two do not need to move together. Firms may expand their overall number of subsidiaries proportionally, raising the count without raising the share. In that case it remains unclear whether this results would be driven by the introduction of substance rules. Observing both outcomes increase instead would tell us that the effect is driven by substance rules making low-tax subsidiaries more attractive again and would suggest expansion or reallocation of subsidiaries or a combination of the two.

However, whether multinationals shift toward low-tax jurisdictions depends on whether the expected tax benefits exceed the costs of creating and maintaining sufficient substance. Establishing and substantiating a subsidiary involves administrative expenses, compliance costs, and the need for real business operations, as well as reputational concerns (Scholes and Wolfson, 1992; Graham et al., 2014; Gallemore et al., 2014; Schanz et al., 2025). When these costs are relatively high, firms may choose not to establish low-tax subsidiaries even when substance-based exemptions are available.

We further expect the response to concentrate among firms for which the exemption relaxes the tightest constraint. Multinationals headquartered in countries whose CFC rules apply a high tax rate threshold face the tighter constraint before the reform, since a broader set of low-tax jurisdictions falls within the scope of their CFC rules. For these firms, the exemption opens an avenue that was previously not available. A second question is whether firms differ in their response across firm size and network structure. Here the direction is less clear ex ante. Larger and more widely dispersed multinationals have greater capacity to establish and substantiate a low-tax subsidiary, but they are also more likely to operate in low-tax jurisdictions already or to have other income shifting arrangements in place, so the exemption may represent a smaller change in their opportunity set (Rego, 2003; Balakrishnan et al., 2019; Francois and Vicard, 2025). Smaller multinationals with narrower networks gain access to a route that was previously out of reach, though establishing a qualifying subsidiary still requires resources they may lack. We examine these questions in our cross-sectional

analysis.

The Cadbury Schweppes ruling provides an ideal setting to test Hypothesis 1. Prior to the ruling, EU countries with CFC regimes in place did not provide for substance-based exemptions. As a result, multinationals headquartered in these countries had limited incentives to use subsidiaries in low-tax countries for income shifting, since passive income earned by such subsidiaries was subject to CFC taxation at their parent companies' higher tax rate. Consequently, subsidiary locations in low-tax countries were likely driven primarily by economic considerations. Following the ruling, all EU countries with CFC regimes introduced substance-based exemptions, thereby creating incentives for multinationals to establish subsidiaries in low-tax countries for income shifting purposes, provided that these subsidiaries satisfy the substance requirements.

Our second hypothesis examines how firms adjust economic activity within existing low-tax subsidiaries when CFC rules with substance exemptions are introduced. Prior literature documents that firms respond to anti-tax avoidance regulations by reducing investment and employment in low-tax entities (Suárez-Serrato, 2019; De Mooij and Liu, 2021; Bilicka et al., 2022). Substance-based exemptions may alter these responses. Rather than reducing activity, firms can preserve the tax advantages of low-tax locations by demonstrating sufficient economic activity, and prior work in other settings finds that substantiating operations enables firms to retain the benefits of reporting income in low-tax countries (Williams, 2018; De Simone and Olbert, 2022; Drake et al., 2022). We therefore expect firms to respond to substance-based exemptions by increasing economic activity in affected subsidiaries:

Hypothesis 2: The introduction of CFC provisions with substance rules increases economic activity in low-tax subsidiaries.

How firms meet substance requirements is less clear. As described in Section 2.1, substance requirements under CFC regimes specify that firms maintain "staff, equipment, assets, and premises" and perform meaningful people functions (European Union, 2016), without establishing precise thresholds, which leaves firms considerable scope in how they respond. One possibility is that firms respond as intended by building up real economic activity in the subsidiary (Williams, 2018; De Simone and Olbert, 2022; Drake et al., 2022). Because the definition places particular weight on qualified personnel rather than on physical capital, we expect an increase to show up in employment more than in investment. Another possibility is that firms rely on reporting-based adjustments, for instance by allocating existing employees to low-tax affiliates without physical relocation through remote work or shared service arrangements, or by employing or reallocating higher-paid personnel who perform the decision-making functions the requirement emphasizes (Chen et al., 2023; Bornemann et al., 2023; Cabral et al., 2025).¹⁰ Both responses would raise personnel costs by more than headcount,

¹⁰As noted in Section 2.1, service providers actively advertise such arrangements to multinational

since each involves adding higher-paid staff rather than a broad expansion in workforce. This gap should be widest under a reporting-based adjustment where costs shift to the subsidiary with little or no change in the number of people employed. We cannot fully disentangle the two, though a large increase in personnel costs relative to headcount is more consistent with a reporting response.

Even where firms build up economic activity, this need not reflect new value creation for the multinational group. Firms could instead shift investment and employment away from related subsidiaries in higher-tax jurisdictions, satisfying the substance requirement by reallocating existing activity rather than expanding it. Prior research documents that low-tax operations can lead to employment declines in higher-tax affiliates as firms relocate activity to meet substance requirements or justify income shifting arrangements (Glaeser et al., 2023; Davies et al., 2026). To assess whether increases in low-tax subsidiaries reflect within-group reallocation, we examine subsidiaries in higher-tax countries that do not qualify as CFCs but belong to the same multinational groups.

We further expect the response to concentrate where firms have the most income at stake and the most adjustment to make. We therefore predict stronger responses among subsidiaries facing larger tax rate differentials relative to their parent. We also expect stronger responses among subsidiaries with a higher share of passive income, since more of their income falls within the scope of CFC taxation, and among subsidiaries with lower pre-existing levels of activity, since these must adjust the most to satisfy the requirement. Finding the response concentrated among these subsidiaries would tie the adjustments we document directly to the substance requirement.

The ATAD setting is well suited to test Hypothesis 2. Several EU countries implemented CFC rules with substance-based exemptions for the first time as part of the ATAD. Prior to implementation, multinationals headquartered in these countries could use low-tax subsidiaries for income shifting purposes without being subject to CFC taxation. Following the introduction of CFC rules, the passive income of low-tax subsidiaries became subject to CFC taxation unless the subsidiaries satisfied the applicable substance requirements, thereby creating incentives for multinationals to increase economic activity within these subsidiaries.

3 Research Design and Data

3.1 Empirics

We test our predictions on firms' responses to substance rules using two difference-in-differences designs that exploit quasi-exogenous variation the introduction of substance-based exemptions in CFC regimes. The two settings are complementary. The Cadbury Schweppes ruling introduced substance-based exemptions for multinationals already

firms.

operating under CFC rules, which lets us study where they choose to locate new subsidiaries. But because these subsidiaries are established only after the ruling, it offers no pre-period against which to measure changes in their economic activity. The ATAD setting overcomes this limitation. By newly subjecting existing low-tax subsidiaries to substance requirements, we can leverage a before-and-after comparison that lets us study how firms adjust economic activity within subsidiaries they already operate. In both settings, identification comes from comparing a treatment group that became subject to substance-based exemptions to a control group that did not, before and after the change. We restrict our analysis to subsidiaries located in the EEA, since substance-based exemptions under both the Cadbury Schweppes ruling and the ATAD apply only to EEA subsidiaries.¹¹

CFC Subsidiaries in the Cadbury Schweppes Setting

We compare the subsidiary location decisions of multinationals affected by the 2006 Cadbury Schweppes ruling to those of unaffected multinationals. The treatment group consists of multinational groups headquartered in the eleven EU countries that had CFC rules in place prior to 2006 and were required to introduce substance-based exemptions following the ruling. Multinational groups headquartered in the 17 EU countries without CFC rules serve as the control group.¹² We designate 2006 as the first treatment year, corresponding to the ECJ ruling in September 2006 ([Ruf and Weichenrieder, 2013](#); [Schenkelberg, 2020](#); [Davies et al., 2026](#)). Because the ruling increased the incentive to locate subsidiaries in low-tax EEA countries, we classify subsidiaries by whether their host country falls within the scope of the parent’s CFC rules. Subsidiaries in low-tax host countries qualify as CFC subsidiaries, while we qualify those in higher-tax host countries fall outside the scope of the rules as non-CFC subsidiaries.

We examine two outcomes that together capture where multinationals locate their subsidiaries. The first is the count of CFC subsidiaries, measured as the number of low-tax subsidiaries a multinational holds in a given year, which reflects its presence in low-tax jurisdictions in absolute terms. The second is the share of CFC subsidiaries relative to the multinational’s total number of subsidiaries, which reflects the composition of its subsidiary network. Estimating both lets us distinguish whether multinationals expand their presence in low-tax jurisdictions, redirect a greater share of their subsidiaries toward them, or both. For multinational group g located in home country m in year t , we estimate:

$$Y_{g,t} = \beta_1 \times (CS\ Treat_m \times CS\ Post_t) + \gamma_1 \times W_{g,t-1} + \gamma_2 \times W_{m,t} + \lambda_g + \omega_t + \varepsilon_{g,m,t} \quad (1)$$

¹¹Restricting our analysis to EEA subsidiaries also improves data coverage, since ownership links to subsidiaries in classic tax havens outside Europe are systematically under-reported in both public and administrative datasets ([Olbert et al., 2024](#)).

¹²Appendix Table A-2 provides an overview of the countries in the treatment and control group.

where $Y_{g,t}$ is the count or share of CFC subsidiaries. $CS\ Treat_m$ equals one for multinational groups headquartered in EU countries with pre-existing CFC rules and zero otherwise. $CS\ Post_t$ equals one from 2006 onward. We include lagged group-level controls ($W_{g,t-1}$) for consolidated leverage, total assets, and the number of employees, and home country controls ($W_{m,t}$) for the statutory CIT rate and GDP (Markle and Robinson, 2012; De Simone and Olbert, 2022). We include multinational group fixed effects (λ_g) to absorb time-invariant differences across groups, and year fixed effects (ω_t) to absorb common time shocks. Standard errors are clustered at the multinational group level.

Because the two outcomes differ in their statistical properties, we estimate each using three specifications rather than relying on a single functional form. For the count of CFC subsidiaries, which is non-negative and contains many zeros, we estimate the equation in logs, using the inverse hyperbolic sine transformation, and by Poisson pseudo-maximum likelihood.¹³ For the share of CFC subsidiaries, which is bounded between zero and one, we estimate a linear specification in levels, a fractional logit specification (Papke and Wooldridge, 1996, 2008), and PPML.¹⁴

The coefficient β_1 captures the differential change for treated relative to control multinationals. Consistent with Hypothesis 1, we expect β_1 to be positive for both the count and share of CFC subsidiaries. Identification relies on the assumption of parallel trends. Specifically, we assume that, absent the ruling, the subsidiary location patterns of treated and control multinationals would have followed the same trend, even if their levels differed. This distinction matters in our setting, since multinationals subject to CFC rules had, before the ruling, a weaker incentive to locate subsidiaries in low-tax jurisdictions and therefore likely started from a lower level. What identification requires is not that the two groups shared the same location patterns, but that the difference between them would have remained stable in the absence of the ruling. We assess the plausibility of this assumption using the event-study estimates in Figure 1, which show no differential pre-trends across specifications. In additional tests, we replace year fixed effects with industry-year fixed effects, and re-estimate the model excluding one host country at a time.

Economic Activity in the ATAD Setting

¹³Not all multinational groups hold a CFC subsidiary in every year, so the count is zero for a substantial share of observations. Therefore we apply a $\log(y + 1)$ transformation. The IHS transformation $\text{asinh}(y) = \log(y + y^2 + 1)$ is defined at zero (Aihounton and Henningsen, 2021). PPML accommodates zeros without transforming the dependent variable by modelling the conditional mean $E[Y|X] = \exp(\alpha + \beta X)$ (Santos Silva and Tenreiro, 2006; Cohn et al., 2022; Chen and Roth, 2024; Wooldridge, 2023). In Poisson regressions, the coefficients are semi-elasticities, so the implied proportional treatment effect is $(\exp(\beta_1) - 1) * 100$.

¹⁴The level specification gives the effect in percentage points. The fractional logit is a quasi-maximum likelihood estimator with a logit link, taking into account the bounds of the outcome and yields coefficients interpretable as likelihoods. The PPML specification measures the proportional change in the share.

We examine how existing subsidiaries adjust economic activity following the introduction of CFC rules with substance-based exemptions under the ATAD. Although the ATAD mandated CFC provisions to enter into force at the end of 2018, firms were informed about the forthcoming rules with the ATAD proposal in January 2016 and its final adoption in July 2016. We therefore designate 2016 as the first treatment year to capture anticipatory adjustments.¹⁵ The treatment group consists of subsidiaries of multinational groups headquartered in thirteen EU countries that newly implemented CFC rules under the ATAD. The control group comprises subsidiaries of groups headquartered in nine EU countries with pre-existing CFC rules.¹⁶ We conduct separate analyses for low-tax CFC subsidiaries and higher-tax non-CFC subsidiaries.

For subsidiary i located in host country s with parent in home country m in year t , we estimate the following PPML regression:¹⁷

$$Y_{i,t} = \beta_1 \times (ATAD\ Treat_m \times ATAD\ Post_t) + \gamma_1 \times W_{i,t-1} + \gamma_2 \times W_{s,t} + \lambda_i + \omega_t + \varepsilon_{i,s,t} \quad (2)$$

$Y_{i,t}$ is unconsolidated fixed assets, number of employees, and personnel costs, each entering our main specification in logs.¹⁸ $ATAD\ Treat_m$ equals one for subsidiaries of groups that became subject to CFC rules with substance-based exemptions for the first time under the ATAD, and zero for subsidiaries of groups already subject to such rules. $ATAD\ Post_t$ equals zero for 2012 to 2015 and one for 2016 to 2019. We include lagged subsidiary controls ($W_{i,t-1}$) and host country controls ($W_{s,t}$) for GDP and the statutory CIT rate. To account for subsidiary size without controlling for the outcome itself, we control for the number of employees when examining fixed assets and for total assets when examining employment outcomes. We include subsidiary fixed effects (λ_i) to absorb time-invariant differences across subsidiaries, and year fixed effects (ω_t) to absorb common time shocks. In alternative specifications, we add host country-year and industry-year fixed effects to absorb time-varying host country characteristics and differential trends across industries. Standard errors are clustered at the subsidiary level.

Consistent with Hypothesis 2, we expect β_1 to be positive for CFC subsidiaries,

¹⁵We support this choice by conducting a media coverage analysis with Factiva, which highlights the spike in newspaper attention throughout 2016 (see Appendix Figure A-3).

¹⁶See Appendix Table A-2 for an overview of the countries in the treatment and control group. We exclude six countries from the analysis due to missing low-tax thresholds (Denmark, Estonia, and Latvia), the introduction of CFC rules during the pre-treatment period (Greece and Poland), or Brexit (United Kingdom). Although some of the countries in the control group amended their existing CFC provisions at the end of 2018, the legislative changes made, e.g., the extension of CFC rules to permanent establishments, are only of minor relevance for the dependent variables considered in our analyses and thus we assume them to have a negligible effect.

¹⁷Poisson regressions are well-suited even when the outcome variable is continuous (Santos Silva and Tenreiro, 2011; Cohn et al., 2022; Wooldridge, 2023). We calculate the estimated treatment effect as $(\exp(\beta_1) - 1) * 100$.

¹⁸Because some subsidiaries report zero fixed assets or employees, we apply a $\log(y+1)$ transformation for these outcomes.

and more pronounced for employment outcomes than for investment, given that the ATAD’s substance requirements place particular weight on qualified personnel rather than on physical capital. For non-CFC subsidiaries, we expect β_1 to be negative if the increase in CFC subsidiaries’ economic activity reflect reallocation within the group. Again, we report three specifications for each outcome. Our primary estimates come from the log specification, which is standard for continuous outcomes of this kind and yields coefficients directly interpretable as proportional changes. We additionally report results for using an IHS and PPML estimation. Identification relies on the parallel trends assumption, namely that, absent the ATAD, economic activity in the subsidiaries of newly treated and previously treated groups would have followed the same trend. We assess the plausibility of this assumption using the event-study estimates in Figure ??, which show no differential pre-trends. In additional tests, we add industry-year fixed effects to absorb trends across industries, re-estimate the model excluding one host country at a time, and apply entropy balancing to improve comparability between treated and control subsidiaries.

Cross-Sectional Tests In both settings we examine where the responses concentrate across firms, which helps to investigate the mechanisms that drive the effects and to establish that the effects are driven by substance requirements rather than by coinciding trends. We implement these tests by splitting each sample into subgroups based on pre-treatment averages of the relevant firm or subsidiary characteristic and re-estimating the baseline specification within each subgroup. In the Cadbury Schweppes setting, we split multinationals by their exposure to the CFC penalty and by their size and organizational structure. In the ATAD setting, we split subsidiaries by their tax rate differential relative to the parent, their share of passive income, and their pre-existing level of economic activity. We describe the construction of each split in the corresponding results sections.

3.2 Sample Construction

We construct separate samples for each setting. For both, we use data for European multinational groups and their subsidiaries from the Orbis database provided by Moody’s. Orbis combines financial statement and industry information with detailed ownership data, allowing us to identify subsidiaries controlled by a parent through direct or indirect majority ownership. We define control based on the relationship between each subsidiary and its global ultimate owner, the highest-ranking shareholder at the top of the ownership chain that holds a majority stake.

Central to our analysis is the classification of subsidiaries as CFC or non-CFC, since this determines which subsidiaries are exposed to CFC rules and thus to the substance requirement. A subsidiary qualifies as a CFC only if it is both controlled by the parent and located in a low-tax country. We classify a host country as low-tax if its statutory

CIT rate falls below the applicable CFC threshold of the multinational's home country, or if it provides a preferential income tax regime (De Simone and Olbert, 2022) or is commonly classified as a tax haven (Bennedsen and Zeume, 2018). Consistent with prior literature, we identify Cyprus, Ireland, Liechtenstein, Luxembourg, Malta, and the Netherlands as EEA jurisdictions that offer preferential tax regimes or are widely regarded as tax havens. We also account for country-specific exemption lists where applicable. Subsidiaries in host countries whose tax rate exceeds the home country's CFC threshold are classified as non-CFC. Table A-2 in the Appendix summarizes the applicable tax thresholds across EU countries for both settings.

In the Cadbury Schweppes setting a complication arises in classifying subsidiaries because whether a host country counts as low-tax depends on the home country's CFC threshold, yet control group countries have no CFC rules and therefore no such threshold. We can thus observe the relevant threshold only for multinationals headquartered in countries with pre-existing CFC regimes, which form the treatment group. To keep the classification comparable across groups, we apply the same set of low-tax and higher-tax host countries to control-group multinationals as we use for the treatment group. In the ATAD setting this issue does not arise, since all EU home countries have CFC regimes in place by the end of the sample period, which lets us determine home-country-specific sets of low-tax host countries for every multinational.

The Cadbury Schweppes sample spans 2002 to 2008 and is constructed at the group-year level, since we examine here where the multinational as a whole locates its subsidiaries. Following prior literature (Clifford, 2019; Schenkelberg, 2020; Prettl, 2017), we rely on an ownership snapshot from 2012, as complete annual ownership histories are unavailable for this earlier period.¹⁹ For each multinational group, we identify all worldwide subsidiaries, their incorporation dates, and the applicable host country tax rates, from which we construct the number of CFC and non-CFC subsidiaries per multinational-year. Starting from a balanced firm-year panel, we drop observations with missing total assets or control variables, require that a group hold at least one EEA subsidiary and at least one CFC subsidiary during the sample period, and drop firms with insufficient coverage.²⁰ This yields 8,035 multinational group-year observations from 1,840 multinational groups, as detailed in Table 1 Panel A.

We additionally construct a subsidiary-level sample for the Cadbury Schweppes setting to examine the economic activity of newly incorporated CFC subsidiaries. We

¹⁹Relying on the 2012 snapshot introduces potential survivorship bias, since subsidiaries that ceased to exist before 2012 are not observed. However, as any subsidiaries missing from the snapshot are unobserved for treated and control groups alike, a level difference of this kind differences out in our design. If anything, a treated multinational that established a low-tax subsidiary in response to the ruling but closed it before 2012 would go unobserved, which would understate rather than inflate our estimated effect.

²⁰Groups that never hold a CFC subsidiary provide no variation in our outcome measures and would not contribute to the estimation.

restrict this Cadbury Schweppes subsidiary sample to subsidiaries incorporated in 2006 or 2007. We use this for conducting descriptive analysis on the economic activity of these newly established subsidiaries in Section 4.4. Its construction is detailed in Table 1 Panel A.

For the ATAD setting, we use a subsidiary-year sample spanning 2012 to 2019, since we examine how individual subsidiaries adjust their economic activity. For this setting, we retrieve annual historic ownership data throughout the entire sample period, allowing us to exclude subsidiaries that experience ownership changes and thus keep our analysis to a stable set of subsidiaries. We focus on CFC and non-CFC subsidiaries located in the EEA and merge unconsolidated financial data and industry classifications. We retain observations with non-missing total assets and control variables. Because our analysis examines how activity adjusts within existing subsidiaries, we restrict the sample to subsidiaries observed at least once in both the pre-treatment period (2012–2015) and the post-treatment period (2016–2019). Table 1 Panel B details the sample selection procedure, yielding xx subsidiary-year observations from xx subsidiaries.

We compile country-level information from several sources. Data on CFC rules are drawn from IBFD, Clifford (2019), and Spengel et al. (2026). Statutory CIT rates are hand-collected from the OECD, the Tax Foundation, and EY.²¹ GDP data are obtained from the World Bank’s World Development Indicators and converted to EUR using annual average exchange rates from Eurostat.²²

3.3 Descriptive Statistics

Table 3 presents descriptive statistics for both samples, with group- and subsidiary-level statistics for the Cadbury Schweppes setting in Panel A and subsidiary-level statistics for the ATAD setting, separately for CFC and non-CFC subsidiaries, in Panel B. All variables are defined in Appendix Table A-1.

In the Cadbury Schweppes group sample, multinationals are large on average, holding 28.9 subsidiaries and EUR 2.6 bn in total assets. The distributions of these variables are highly right-skewed, however. The corresponding medians are 9 subsidiaries and EUR 278 million, indicating that the typical group is considerably smaller. Of a group’s subsidiaries, an average (median) of 4.43 (2) qualify as CFC subsidiaries.²³ The Cadbury Schweppes subsidiary sample, which we use for descriptive evidence on substance in this setting, contains newly incorporated CFC subsidiaries that are small in

²¹OECD Tax Database, https://stats.oecd.org/Index.aspx?DataSetCode=CTS_CIT (05/20/2024); Tax Foundation, <https://taxfoundation.org/data/all/global/corporate-tax-rates-by-county-2024/> (03/10/2022).

²²World Bank, <https://databank.worldbank.org/source/world-development-indicators> (05/20/2024); Eurostat, <https://ec.europa.eu/eurostat/web/exchange-and-interest-rates/database> (05/20/2024).

²³The CFC and non-CFC classifications in our setting apply only to subsidiaries located in the EEA, as substance requirements within CFC rules pertain exclusively to EEA countries.

their first years, holding on average EUR 1.07 million in fixed assets, with a median of EUR 20,000.

In the ATAD sample, CFC and non-CFC subsidiaries differ systematically. CFC subsidiaries are smaller, holding EUR xx million in fixed assets and employing xx workers on average, compared with EUR xx million and xx workers for non-CFC subsidiaries. By construction, they also face lower taxation, with a mean statutory CIT rate of xx% against xx% for non-CFC subsidiaries. The larger number of non-CFC subsidiaries reflects that most subsidiaries operate in countries whose tax rate exceeds the applicable CFC threshold, together with weaker Orbis coverage in low-tax countries, particularly for financial data ([Fuest et al., 2022](#); [Olbert et al., 2024](#)).

4 Effects on CFC Subsidiaries

4.1 Results

Table 4 presents the results of testing Hypothesis 1, which predicts that following the introduction of substance-based exemptions, multinationals subject to CFC rules increase their subsidiary activity in low-tax countries. The intuition is that the Cadbury Schweppes ruling allows multinationals subject to CFC rules to avoid the CFC tax penalty by establishing economic substance in low-tax subsidiaries, making such jurisdictions more attractive for new subsidiary activity relative to control multinationals for whom this constraint does not apply. We examine two outcomes. The count of CFC subsidiaries captures how many low-tax subsidiaries a multinational holds in a given year, while the share of CFC subsidiaries relative to the multinational's total captures the composition of its subsidiary network.

Panel A reports estimates for the count of CFC subsidiaries. The coefficients are positive and statistically significant across all three specifications. Our preferred estimate in Column (2), which includes control variables, implies that treated multinationals increase their number of CFC subsidiaries by 4.6% following the ruling.²⁴ Given a sample mean of 4.43 CFC subsidiaries, this corresponds to roughly 0.20 additional CFC subsidiaries per multinational per year. This magnitude is broadly similar to prior evidence that tax rules shape where multinationals locate subsidiaries ([Barrios et al., 2012](#); [Clifford, 2019](#)).

Panel B reports estimates for the share of CFC subsidiaries. The coefficients are positive and statistically significant across the level, fractional logit, and the Poisson specification. The level estimate in Column (2) implies that treated multinationals increase their share of CFC subsidiaries by 1.23 percentage points, which corresponds to a relative increase of roughly 7.5% against a sample mean share of approximately 16% CFC subsidiaries. The fractional logit specification indicates that the likelihood of a subsidiary being located in a low-tax rather than a higher-tax jurisdiction rises by

²⁴The percentage change is derived as $(\exp(0.0447) - 1) \times 100$.

roughly 10% for treated multinationals relative to control multinationals.²⁵ Taken together, the two panels indicate that treated multinationals do not only add subsidiaries in low-tax jurisdictions, but also redirect a greater share of their subsidiary network toward these jurisdictions.

To shed light on the dynamics of these results and to confirm that the results are not driven by observable pre-existing trend differences between treated and control multinationals, Figure 1 presents event-study estimates, replacing the *Post* indicator with separate *Year* indicators. Panel A shows the count and Panel B the share of CFC subsidiaries, respectively. Across both panels and all specifications (log, IHS, PPML, level, logit), the pre-treatment coefficients are small and indistinguishable from zero, supporting the parallel trends assumption underlying our design. The effects on the count and the share of CFC subsidiaries emerge in the first year following the Cadbury Schweppes ruling and remain positive and significant through the end of the sample period. This pattern indicates that the compositional shift toward low-tax jurisdictions is sustained rather than a one-time adjustment.

4.2 Robustness and Plausibility Tests

While the observed patterns are consistent with multinationals responding to substance rules by shifting subsidiaries to low-tax countries, we next consider alternative explanations. First, industry-specific time trends, rather than the substance-based exemption introduced by the Cadbury Schweppes ruling, could drive the observed increase in CFC subsidiaries among treated multinationals. To assess this, we re-estimate our specifications replacing year fixed effects with industry-year fixed effects, thereby absorbing any differential trends across industries. Appendix Table A-4 reports the results. The coefficients remain statistically significant and similar in magnitude to the baseline estimates, indicating that differential trends across industries do not explain our findings.

Second, we examine whether treated multinationals also change their subsidiary decisions in higher-tax countries that fall outside the scope of CFC rules. If the response is specific to the substance requirement and not driven by an overall expansion of their subsidiary networks, we should observe no corresponding increase in higher-tax countries. If instead the increase in the count of CFC subsidiaries reflects reallocation within the subsidiary network, we would expect a simultaneous decline in the count of non-CFC subsidiaries. Appendix Table A-5 reports the DiD results for non-CFC subsidiaries. Panel A shows the results on the number and Panel B on the share of non-CFC subsidiaries. We mirror the same specifications as for the CFC subsidiaries.²⁶ The share of non-CFC subsidiaries declines across all specifications and is

²⁵This likelihood is derived as $\exp(0.0943)$.

²⁶We include log, IHS, and PPML for the count outcome, and level, fractional logit and PPML for the share outcome.

statistically significant in the level and fractional logit specifications, mirroring the increase we document for the CFC share and consistent with a compositional shift within the subsidiary network toward low-tax subsidiaries. For the count of non-CFC subsidiaries, the estimates are positive and significant in the log and IHS specifications. This would speak against our results and lean toward an overall network expansions of treated vs. control multinationals. However, when we look at the corresponding event study results in Appendix Figure A-1 they show negative pre-treatment trends in these specifications, indicating that the positive post-period coefficients reflect a pre-existing divergence in the number of non-CFC subsidiaries between treated and control multinationals rather than a response to the ruling. The Poisson specification, which shows no such pre-trends, yields no significant effect on the non-CFC count. Taken together, these results indicate that the response is specific to low-tax jurisdictions rather than reflecting a broader change in subsidiary activity.²⁷

Third, we assess whether our results are driven by multinationals headquartered in any single country by implementing a leave-one-out design that sequentially excludes each home country from the sample. Appendix Figure A-2 presents the resulting coefficient estimates for the count of CFC subsidiaries in Panel (A) and the share in Panel (B). The estimates remain statistically significant and of comparable magnitude across all country exclusions, confirming that no individual country drives the baseline results and that our findings reflect a broad-based response among treated multinationals.

4.3 Cross-Sectional Evidence

To test the mechanism underlying our main results, we next examine whether the response to the Cadbury Schweppes ruling varies systematically across multinationals. If the substance-based exemption induces multinationals to increase their presence in low-tax countries because it removes the CFC tax penalty for subsidiaries that establish sufficient economic substance, the response should be strongest among multinationals with the most to gain from doing so, or the fewest alternative routes available to them. We therefore examine whether the effects we document are concentrated among multinationals with the strongest incentive or opportunity to shift their presence toward low-tax jurisdictions.

First, a multinational headquartered in a country with a stricter, that is higher, CFC tax threshold has a greater incentive to avoid low-tax jurisdictions before the exemption, because a broader range of foreign tax rates falls under the penalty. For example, a German multinational with an Irish subsidiary would be treated because Ireland's 12.5% rate falls below the German threshold, whereas the same subsidiary held by a Hungarian multinational would not, since Hungary's threshold is lower at 10%. After

²⁷The increasing number of non-CFC subsidiaries in the treated group that becomes stagnant after the reform even speaks to treated groups changing their subsidiary location patterns after the reform and thus confirms our interpretations.

the introduction of the substance requirement, the German firm therefore has a higher incentive to establish or relocate subsidiaries to jurisdictions such as Ireland, because it can now apply the substance exemption and benefit from the lower tax rate. We test this by splitting our treatment group into multinationals headquartered in countries with a high versus low CFC tax threshold.²⁸ Table 6 Columns (1) and (2) report the results, with Panel A presenting estimates for the count of CFC subsidiaries and Panel B for the share. Consistent with our prediction, both the count and the share increase significantly for multinationals from high-threshold countries, while the corresponding estimates for low-threshold countries are smaller and insignificant.

Second, we examine whether the response varies with a multinational's size and the breadth of its subsidiary network. As discussed in Section 2.2, the expected direction is ambiguous. Larger and more widely dispersed multinationals have greater capacity to establish and substantiate a low-tax subsidiary, but they may also already operate in such jurisdictions or have other income shifting arrangements in place. We measure size by pre-treatment average total assets and network breadth by the pre-treatment number of subsidiaries, splitting the sample at the median of each. Columns (3) and (4) report results by size. Both the count and the share of CFC subsidiaries increase significantly among multinationals with below-median total assets, while the estimates for larger multinationals are smaller and insignificant. Columns (5) and (6) report results by the number of pre-existing subsidiaries and show the same pattern. Both outcomes respond among multinationals with below-median subsidiary counts, with no significant response among those operating broader networks. The response is therefore concentrated among multinationals that were smaller and less widely dispersed before the reform, consistent with the substance rule opening access to low-tax jurisdictions for firms that previously had few routes to them. Larger multinationals with extensive networks, by contrast, show no significant response, consistent with the exemption representing a smaller change in their opportunity set, in line with prior work finding that larger and more complex multinationals already have income shifting arrangements available to them (Rego, 2003; Balakrishnan et al., 2019; Francois and Vicard, 2025).

Taken together, our results suggest that the Cadbury Schweppes ruling led treated multinationals both to add subsidiaries in low-tax jurisdictions and to redirect a greater share of their network toward them. Unlike traditional CFC rules, which prior literature shows induce firms to close or relocate subsidiaries in low-tax countries (Voget, 2011; Clifford, 2019), we find that substance-based exemptions increase the attractiveness of these jurisdictions by providing a route to their tax advantages while avoiding CFC taxation, and that firms respond precisely where the exemption represents the

²⁸Countries with a high CFC tax threshold include Denmark (21%), Germany (25%), Spain (26%), the United Kingdom (22.5%), and France (17.48%). Countries with a low threshold include Finland (15.6%), Hungary (10.67%), Lithuania (11.25%), Portugal (15%), and Sweden (15.4%).

largest change in their opportunity set.

4.4 Motivational Evidence on Substance

Our findings imply that substance-based exemptions induce multinationals to establish subsidiaries in low-tax countries. For this interpretation to hold, newly established subsidiaries must exhibit sufficient economic activity to qualify for the substance-based exemption and thus fall outside the scope of CFC taxation. Ideally, we would examine whether these newly incorporated subsidiaries have more investment and employment using a difference-in-differences design, but we cannot do so because these subsidiaries are incorporated only after the ruling, leaving no pre-period for identification. Nonetheless, we provide motivational evidence by descriptively comparing investment and employment of newly incorporated CFC subsidiaries belonging to treated and untreated multinationals in the years following their establishment. If treated multinationals establish subsidiaries with sufficient economic activity to meet the substance threshold and benefit from the exemption, we would expect their newly incorporated CFC subsidiaries to exhibit systematically higher investment and employment relative to those of control multinationals.

Table 7 reports regressions of fixed assets (Panel A), number of employees (Panel B), and personnel costs (Panel C) for CFC subsidiaries incorporated after the 2006 ruling on an indicator for whether the parent multinational is subject to substance-based CFC rules.²⁹ Column (1) presents results without controls, and Column (2) adds firm-level controls. Both include host country–year fixed effects, comparing subsidiaries operating in the same host country and year but belonging to different multinational groups.³⁰ Column (3) replaces host country–year fixed effects with host country–age fixed effects, comparing subsidiaries within the same country at the same age after incorporation. Column (4) includes host country and industry–year fixed effects, allowing for differential trends across industries.

For fixed assets in Panel A the coefficients on the treatment indicator are positive and statistically significant across all four specifications. The estimate in Column (2) implies that newly incorporated subsidiaries of treated multinationals hold substantially more fixed assets than those of control multinationals, consistent with treated multinationals building up physical capital in these subsidiaries to meet the substance threshold.³¹

²⁹We restrict the sample to CFC subsidiaries incorporated after the ruling in 2006 and 2007 to ensure a sufficient number of observations. This restriction is further supported by the event-study evidence, which shows that the increase in the number of subsidiaries is concentrated in the first two years following the ruling.

³⁰Note that coefficients on GDP and CIT are not reported in Column (2) as they are perfectly collinear with the host country–year fixed effects and are therefore absorbed.

³¹The implied percentage change in the outcome is derived as $(\exp(1.1390) - 1) * 100$. However, note that this magnitude should be interpreted with caution, since the $\log(y+1)$ transformation approximates a proportional change only when the outcome is large relative to the added constant, and a substantial share of newly incorporated subsidiaries report little or no fixed assets in their first years. The median

The coefficients for the number of employees in Panel B are positive but lack statistical significance once control variables are added. For personnel costs in Panel C, the coefficients on the treatment indicator are positive and statistically significant across all specifications. The stronger difference for personnel costs relative to the number of employees suggests that treated subsidiaries employ higher-paid workers rather than simply more of them, which is consistent with substance requirements emphasizing meaningful people functions.

Taken together, these findings descriptively document a positive association between substance rules in CFC provisions and higher economic activity in newly established low-tax subsidiaries. This pattern is consistent with treated multinationals establishing subsidiaries with sufficient substance to meet the exemption threshold, which would allow them to benefit from the ruling while avoiding CFC taxation on the income generated. We note, however, that these comparisons are descriptive and do not establish a causal effect of the ruling.

5 Effects on Economic Activity

5.1 Results

Having established that substance-based exemptions lead multinationals to shift the composition of their subsidiary activity toward low-tax jurisdictions, we now turn to whether firms respond by building up real economic activity within these subsidiaries. Answering this question requires observing the same subsidiary before and after it becomes subject to substance requirements, which the Cadbury Schweppes setting cannot provide because the newly incorporated subsidiaries we study there have no pre-period. We therefore turn to the introduction of the ATAD, which subjected previously unaffected multinationals to CFC rules with substance-based exemptions for the first time. This setting offers two key advantages. First, because it applies to subsidiaries that already existed before treatment, it provides a pre-period that allows us to estimate the effect on economic activity in a difference-in-differences design rather than relying on descriptive comparisons. Second, by newly subjecting firms to substance requirements, it isolates how multinationals adjust activity in order to preserve the tax advantages of their low-tax subsidiaries. To retain these advantages under the substance requirements, affected firms need to demonstrate sufficient economic substance in the subsidiary. Otherwise they face taxation at the higher parent-country rate. This allows us to test Hypothesis 2 on whether firms increase economic activity within low-tax subsidiaries and to examine the specific margins along which they respond. We consider two distinctions. The first is between investment and employment, which reveals whether firms adjust physical capital, labor, or both. The second

newly incorporated subsidiary holds EUR 20,000 in fixed assets against a sample mean of EUR 1.07 million.

is within employment, between adding headcount and increasing the personnel costs, which reveals whether firms expand their actual workforce or primarily raise labor costs.

Table ?? reports the main estimates from Equation 2. Columns (1) and (2) present results without and with control variables, respectively. For fixed assets (Panel A), the interaction coefficients are negative and statistically insignificant, indicating that affected multinationals do not respond by expanding physical investment. By contrast, the coefficients for the number of employees (Panel B) and personnel costs (Panel C) are positive and statistically significant. The estimates in Column (2) imply that employment increases by xx% and personnel costs rise by xx% in affected CFC subsidiaries. Given sample means of xx employees and EUR xx million in personnel costs, these correspond to increases of approximately xx employees and EUR xx in labor expenses.

Taken together, these estimates support Hypothesis 2 that affected CFC subsidiaries increase economic activity to meet substance requirements and avoid CFC taxation. The results point to two patterns worth examining more closely. First, firms respond through employment rather than investment, adjusting labor while leaving physical capital unchanged. This is consistent with prior evidence that labor is more mobile and less costly to adjust than physical capital (Giroud and Rauh, 2019; De Simone and Olbert, 2022). The imprecise definition of substance in CFC provisions, which specifies neither precise thresholds nor required types of activity, may further ease adjustment along labor-related margins. Second, within employment, the effect on personnel costs is more pronounced than the effect on headcount. As discussed in Section 2.2, this gap is consistent with two mechanisms. Firms may shift the composition of their workforce toward higher-paid employees who perform decision-making functions emphasized in substance requirements, in line with evidence from other activity-based tax regimes (Chen et al., 2023; Bornemann et al., 2023). Alternatively, firms may satisfy substance requirements through reporting adjustments that allocate personnel to low-tax affiliates without a proportional increase in the actual workforce. We cannot fully disentangle these two channels, but both indicate that firms respond to substance rules through targeted labor-related adjustments rather than broad operational changes.

Figure ?? presents event-study estimates that allow us to assess the dynamics of these effects and to test the parallel trends assumption underlying our design. Panels (A), (B), and (C) report the dynamic effects for CFC subsidiaries for fixed assets, number of employees, and personnel costs, respectively.³² Across all three outcomes, the pre-treatment coefficients are small and statistically insignificant, supporting the assumption that treated and control subsidiaries followed parallel trends before the reform. In the post-treatment period, the estimates mirror the patterns from the main

³²The event-study specification replaces the *Post* indicator with separate year indicators that are interacted with the treatment status.

regressions. Fixed assets show no significant response, confirming that affected firms do not adjust physical investment. The number of employees increases significantly in 2016, the first year after treatment, but reverts to insignificance thereafter, whereas personnel costs exhibit a significant increase that begins in 2016 and persists through the end of the sample period. This divergence reinforces the interpretation from the main results. The transitory response in headcount alongside a sustained increase in personnel costs is consistent with firms adjusting the composition of their workforce toward higher-paid roles or reallocating reported personnel, rather than expanding the number of people they employ.

The results so far show that affected subsidiaries increase labor-related activity. A natural question is whether this reflects a genuine expansion of activity or a reallocation of existing resources from elsewhere in the multinational group. If firms are shifting activity rather than creating it, we would expect to see offsetting declines in their subsidiaries located in higher-tax countries, which fall outside the scope of CFC rules and therefore face no substance requirement. To test this, we examine non-CFC subsidiaries belonging to the same parent companies but located in higher-tax jurisdictions.

Table ?? reports the estimates. The interaction coefficients in Columns (1) and (2) are negative across all outcomes, with the employment measures statistically significant at the 1% level. The magnitudes are comparable to those for CFC subsidiaries, implying declines of xx% in headcount and xx% in personnel costs. Figure ?? confirms these patterns dynamically. Panels (A), (B), and (C) display the dynamic effects for non-CFC subsidiaries for fixed assets, number of employees, and personnel costs, respectively. The pre-treatment coefficients are close to zero and statistically insignificant, supporting the parallel trends assumption, while in the post-treatment period the employment-related outcomes exhibit notable declines. Taken together, these results indicate that firms reallocate labor-related activity from higher-tax to lower-tax affiliates rather than expanding it, reinforcing the interpretation that the response to substance requirements operates through shifting existing resources. A direct policy implication is that countries introducing CFC rules with substance-based exemptions may see local employment decline, as multinationals move activity into low-tax jurisdictions to satisfy substance thresholds elsewhere.

5.2 Robustness Tests

We conduct three tests to assess the robustness of our findings. First, our baseline specification could be confounded by time-varying shocks that affect particular host countries or industries, such as local labor market conditions or sector-wide trends in employment. To address this, we re-estimate the baseline specification replacing year fixed effects with host country-year and industry-year fixed effects, which absorb any such shocks common to firms in the same country-year or industry-year. Columns (3)

and (4) of Tables ?? and ?? report the results. For CFC subsidiaries, the employment coefficients remain positive and mostly statistically significant, and for non-CFC subsidiaries they remain significantly negative. These patterns confirm that unobserved host country or industry trends do not drive our findings.

Second, our estimates could be disproportionately influenced by subsidiaries located in any single host country, particularly given that some jurisdictions host a large share of affected subsidiaries. To assess this, we conduct a leave-one-out analysis for CFC subsidiaries that sequentially excludes each host country from the sample. Appendix Figure ?? presents the results for fixed assets (Panel A), number of employees (Panel B), and personnel costs (Panel C). The estimates remain similar in magnitude across all exclusions, indicating that no single host country drives our results.

Third, treated and control subsidiaries may differ systematically in observable characteristics that also shape their economic activity, which could bias a simple comparison between them. To address this, we apply entropy balancing, re-estimating the baseline specification on a sample reweighted to match treated and control subsidiaries on the first three moments of their pre-treatment control variables.³³ Appendix Tables ?? and ?? report the results. For CFC subsidiaries, the coefficient on personnel costs remains economically similar and statistically significant at the 1% level, while the coefficients on fixed assets and headcount are positive but insignificant. For non-CFC subsidiaries, the employment coefficients remain negative and statistically significant. These results confirm that observable pre-treatment differences between groups do not explain our findings.

5.3 Cross-Sectional Evidence

If firms increase economic activity specifically to satisfy substance requirements, the response should be strongest among subsidiaries with the most to gain from qualifying for the exemption. We therefore examine whether the effects are concentrated among subsidiaries with the strongest incentives to adjust. A subsidiary facing a large tax rate differential relative to its parent company would incur a higher CFC tax penalty if it does not build up substance. The same holds for a subsidiary with a high share of passive income, which has more income exposed to CFC taxation. Finally, a subsidiary that maintains only a low pre-existing level of substantive activity may need to adjust more to fall outside the scope of CFC taxation. We therefore partition the sample along three dimensions: (i) the tax rate differential between the host and home countries, (ii) the share of passive income, and (iii) the pre-existing level of economic substance. Table 16 reports the results, with Panel A presenting estimates for fixed assets, Panel B for number of employees, and Panel C for personnel costs.

³³Entropy balancing reweights control observations so that their weighted distribution of covariates matches that of the treated group on the mean, variance, and skewness. This ensures that treated and control subsidiaries are comparable in their observable pre-treatment characteristics without discarding observations.

Columns (1) and (2) report results for subsamples with above- and below-median tax rate differentials between the host and home countries. The tax saving from keeping income in the low-tax subsidiary, and therefore the incentive to meet the substance requirement, is larger when this differential is wide. Consistent with this, the employment coefficients are significantly positive for subsidiaries facing high tax differentials, while the estimates for subsidiaries with smaller differentials are not statistically significant.

Columns (3) and (4) present estimates for subsamples defined by above- and below-median shares of passive income.³⁴ Since passive income is the type of income that CFC rules attribute back to the parent when substance is lacking, subsidiaries with higher passive income shares have more at stake and stronger incentives to qualify for the exemption. Consistent with this prediction, subsidiaries with high passive income shares exhibit significant increases in employment outcomes, whereas those with low passive income shares do not.

Columns (5) and (6) report results for subsamples defined by pre-existing levels of economic activity. We classify subsidiaries as low-substance if their pre-treatment average total assets and number of employees are both below the sample median, and as high-substance if both exceed the median.³⁵ Subsidiaries with low initial substance have the furthest to go to meet the substance requirement and therefore need to expand. Consistent with this, we find that low-substance subsidiaries exhibit significant increases in employment and personnel costs, with coefficients exceeding those in the baseline specification (Table ??), while the estimates for high-substance subsidiaries are smaller and not consistently significant.

Across all three cross-sectional tests, we find no heterogeneous effects on fixed assets, reinforcing that firms respond through labor-related margins rather than investment. More importantly, the employment response is concentrated among subsidiaries with the strongest incentives to qualify for the exemption, those facing large tax differentials, holding more passive income, or starting from low levels of substance. This pattern ties the adjustments we document directly to the substance requirements, since firms respond precisely where qualifying for the exemption is most valuable. It also indicates that firms adjust selectively rather than across the board, increasing labor only where the tax benefit of doing so is largest.

6 Conclusion

We examine how multinational firms respond to substance rules embedded in anti-tax avoidance regulations. Despite the widespread adoption of substance-based ex-

³⁴We measure the passive income share as the pre-treatment average ratio of financial revenue to total revenue.

³⁵Subsidiaries with mixed classifications, with one measure above and one below the median, are excluded from this analysis.

emptions as a central guardrail of modern international tax policy, empirical evidence on how firms respond to these provisions has been limited. Using two regulatory changes to CFC provisions in the EU, we provide evidence on firm responses along two dimensions, where firms locate their subsidiaries and how they adjust economic activity within them.

We first study whether substance-based exemptions affect where firms locate their subsidiaries. Following the Cadbury Schweppes ruling, which required EU countries with existing CFC regimes to introduce substance exemptions, affected multinationals shift the composition of their subsidiaries toward low-tax jurisdictions, increasing the share of both their subsidiaries and their new incorporations located there, with no corresponding change in the counts and no effect in higher-tax countries. This response suggests that substance-based exemptions increase the attractiveness of low-tax jurisdictions by providing a pathway to obtain tax advantages while avoiding CFC taxation. Cross-sectional evidence indicates that this shift is concentrated among the firms with the strongest incentive to use the exemption and the greatest flexibility to reallocate within their existing structure.

We next examine how firms adjust economic activity within existing low-tax subsidiaries once they become subject to CFC rules with substance exemptions. Firms respond primarily through employment rather than investment. While the number of employees increases modestly, personnel costs increase substantially and persistently. This pattern is consistent with labor being more mobile and less costly to adjust than physical capital ([Slemrod, 1992](#); [Giroud and Rauh, 2019](#); [De Simone and Olbert, 2022](#)). The more pronounced effect on personnel costs than on headcount suggests that firms satisfy substance requirements through targeted labor-related adjustments rather than by expanding their real workforce. Moreover, we document offsetting declines in employment and personnel costs among related subsidiaries in higher-tax countries, indicating that the observed increases reflect within-group reallocations and thus come at the expense of activity in higher-tax locations.

Taken together, our evidence points more toward a reallocation of activity to satisfy substance requirements than toward the creation of real new activity, and, within this reallocation, toward adjustments in reported personnel rather than real operational change. We note, however, that our data do not allow us to fully disentangle a shift in real activity from a shift in its reporting. Given that substance definitions in CFC regimes are often vague, typically referring to staff, assets, and premises without specifying precise thresholds, we cannot directly observe whether firms satisfy the legal substance requirements and instead rely on observable proxies. Our findings thus speak to firm behavior under loosely defined substance requirements. Future research examining regimes with more precisely defined thresholds may reveal whether clearer definitions constrain firms' ability to comply through low-cost adjustments, and can

aim to better separate real responses from reporting ones.

Overall, our results have important implications for the design of anti-tax avoidance policies. Firms respond to substance-based exemptions primarily through employment-related adjustments, and appear to do so at low cost, which suggests that imprecise substance definitions may allow firms to retain the tax advantages of low-tax locations without meaningful operational change such as new capital investment. At the same time, the reallocation of employment from higher-tax to low-tax jurisdictions suggests that substance rules can carry unintended consequences for higher-tax countries, including not only reduced tax revenues but also declines in local economic activity.

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Tables and Figures

Table 1: Sample Selection

	N	N Firms
Panel A: Cadbury Schweppes Setting		
<i>Group Sample</i>		
Balanced firm-year panel from Orbis 2002 to 2009 (GUO 50 firms)	15,877,696	1,984,712
Less observations with missing total assets	-15,787,062	-1,965,799
Less observations with missing data to construct control variables	-34,063	-6,035
Less firms without EEA subsidiaries	-9,280	-1,832
Less firms without CFC subsidiary	-36,993	-8,710
Less firms with insufficient coverage	-496	-496
	9,802	1,840
<i>Subsidiary Sample</i>		
Balanced firm-year panel from Orbis 2002 to 2009	24,562,064	3,070,258
Less observations with missing total assets	-18,056,666	-1,769,644
Less firms not incorporated in 2006 or 2007	-3,586,703	-208,592
Less observations before 2006	-2,309,067	-804,551
Less firms not meeting CFC requirements	-460,626	-218,841
Less observations with missing data to construct control variables	-135,894	-59,562
	13,108	9,068
Panel B: ATAD Sample		
Balanced firm-year panel from Orbis 2010 to 2021 (all subsidiary firms)	178,017,456	14,834,788
Less firms missing total assets	- 138,434,316	-11,536,193
Less firms with missing ownership data and ownership changes	-18,349,174	-1,113,204
Less observations for years 2010, 2011, 2020, 2021 (needed for ownership interpolation & lagged variables)	-6,685,645	-71,010
Less firms with insufficient coverage	-4,106,973	-528,477
Less firms not in EEA	-305,834	-45,691
Less firms not classifiable as CFC or non-CFC subsidiary	-9,389,385	-1,433,261
Less observations with missing data to construct control variables	- 233,017	-19,774
Less observations with missing outcome variables	-313,430	-46,274
<i>CFC Subsidiaries</i>	37,929	9,044
<i>Non-CFC Subsidiaries</i>	161,753	31,860
<i>Non-CFC Subsidiaries aggregated to parent level</i>	25,012	3,719

Notes: The table displays the sample selection procedure for the multinationals and subsidiaries considered in the Cadbury Schweppes setting (Panel A), and the subsidiaries considered in the ATAD setting (Panel B). N denotes the number of observations and N Firms denotes the number of firms.

Table 2: Sample Selection: ATAD Setting

	N	N Firms
ATAD Sample		
Balanced firm-year panel from Orbis 2010 to 2021 (all subsidiary firms)	178,017,456	14,834,788
Less firms missing total assets	-138,434,316	-11,536,193
Less firms with missing ownership data and ownership changes	-18,349,174	-1,113,204
Less observations for years 2010, 2011, 2021 (needed for ownership interpolation & lagged variables)	-5,417,511	-65,171
Less firms without pre- and post-treatment year availability	-1,663,971	-475,997
Less firms with insufficient coverage	-2,443,002	-52,480
Less observations for year 2020	-1,268,134	-5,839
Less firms not in EEA	-305,834	-45,691
CFC Subsidiaries		
Less observations not classifiable as treated/control	-9,884,195	-1,503,226
Less observations with missing total assets, employees, GDP, or CIT	-126,029	-11,505
Less observations not meeting the FA = NE = CE alignment (V4)	-87,361	-16,438
	37,929	9,044
Of which treated	1,303	
Of which control	7,741	
Non-CFC Subsidiaries		
Less observations not classifiable as treated/control	-3,539,907	-538,255
Less subsidiaries whose parent group's CFC subsidiary is not in the desc	-6,100,797	-931,993
Less observations with missing total assets, employees, GDP, or CIT	-106,988	-8,269
Less observations not meeting the FA = NE = CE alignment (V4)	-226,069	-29,836
	161,753	31,860
Of which treated	7,645	
Of which control	24,215	
Non-CFC Subsidiaries aggregated to parent level	25,012	3,719

Notes: This table displays summary statistics for the ATAD setting. Panel A shows the summary statistics for the CFC subsidiary sample and Panel B for the non-CFC subsidiary sample.

Table 3: Summary Statistics

	N	Mean	SD	P25	P50	P75
Panel A: Cadbury Schweppes Setting						
<i>Group Sample</i>						
Number Total Subs. _{g,t}	8,035	28.90	60.98	3.00	9.00	26.00
Number Total EEA Subs. _{g,t}	8,035	26.59	57.60	3.00	9.00	24.00
Number CFC Subs. _{g,t}	8,035	4.43	8.38	1.00	2.00	4.00
Number Non-CFC Subs. _{g,t}	8,035	22.16	53.02	1.00	6.00	19.00
Total Assets _{$g,t-1$} (in m)	8,035	2,616.30	5,434.57	66.47	278.76	1,693.67
Number of Employees _{$g,t-1$}	8,035	9,462.11	18,213.31	313.50	1,427.00	7,463.00
Leverage _{$g,t-1$} (in %)	8,035	16.14	15.48	3.63	12.59	23.82
GDP _{m,t} (in b)	8,035	1,136.96	883.44	292.38	844.39	2,089.80
CIT _{m,t} (in %)	8,035	30.34	6.39	28.00	30.00	34.95
<i>Subsidiary Sample</i>						
Fixed Assets _{i,t} (in m)	13,108	1.070	6.400	0.000	0.020	0.140
Number of Employees _{i,t}	13,108	17.330	54.290	2.000	5.000	13.000
Costs of Employees _{i,t}	13,108	187,993.530	1,003,773.280	3,100.000	18,408.000	76,000.000
Total Assets _{i,t} (in m)	13,108	2.260	12.600	0.020	0.130	0.570
Leverage _{i,t}	13,108	8.040	21.930	0.000	0.000	0.000
GDP _{s,t} (in b)	13,108	111.100	82.540	33.560	116.020	121.010
CIT _{s,t} (in %)	13,108	16.130	4.560	10.000	16.000	19.000
	N	Mean	SD	P25	P50	P75
Panel B: ATAD Setting						
<i>CFC Subsidiary Sample</i>						
<i>Non-CFC Subsidiary Sample</i>						

Notes: The table displays the summary statistics for the group and subsidiary sample considered in the Cadbury Schweppes setting (Panel A), and the subsidiary sample considered in the ATAD setting (Panel B). The subsidiary observations in Panel B are split into subsidiaries classified as CFCs and non-CFCs. Total assets and fixed assets are denoted in thousand EUR. GDP is denoted in million EUR.

Table 4: Effects on the Count and Share of CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Count of CFC Subsidiaries _{g,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	0.0407*** (2.7508)	0.0447*** (2.9230)	0.0551*** (2.9822)	0.0598*** (3.1094)	0.0239 (1.0859)	0.0387* (1.8461)
<i>Log(Total Assets_{g,t-1})</i>		0.0704*** (3.9416)		0.0902*** (3.9168)		0.0880*** (3.8627)
<i>Log(Num. Empl._{g,t-1} + 1)</i>		-0.0070 (-0.7706)		-0.0099 (-0.8437)		-0.0110 (-0.7999)
<i>Leverage_{g,t-1}</i>		0.0006 (1.4598)		0.0007 (1.3683)		0.0008 (1.5088)
<i>Log(GDP_{m,t})</i>		-0.0121 (-0.1082)		-0.0306 (-0.2157)		-0.1953 (-1.5562)
<i>CIT_{m,t}</i>		-0.0015 (-1.1688)		-0.0019 (-1.1858)		-0.0013 (-0.7604)
N	8,035	8,035	8,035	8,035	8,035	8,035
Adj. R ²	0.957	0.958	0.954	0.954		
Pseudo R ²					0.718	0.718
Panel B	Share of CFC Subsidiaries _{g,t}					
	Level	Level	Logit	Logit	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	0.0113** (2.1611)	0.0123** (2.2507)	0.0926** (2.1250)	0.0943** (2.1996)	0.0709*** (4.1589)	0.0678*** (3.8341)
<i>Log(Total Assets_{g,t-1})</i>		0.0089 (1.4999)		0.0872* (1.6787)		0.0265 (1.4383)
<i>Log(Num. Empl._{g,t-1} + 1)</i>		-0.0057* (-1.7088)		-0.0602* (-1.7888)		-0.0176 (-1.5041)
<i>Leverage_{g,t-1}</i>		0.0001 (0.4530)		0.0006 (0.4032)		0.0003 (0.7520)
<i>Log(GDP_{m,t})</i>		0.0200 (0.5229)		0.2371 (0.6361)		-0.0755 (-0.8936)
<i>CIT_{m,t}</i>		-0.0001 (-0.2166)		-0.0001 (-0.0124)		0.0009 (0.8476)
N	8,035	8,035	6,560	6,560	8,035	8,035
Adj. R ²	0.969	0.969				
Pseudo R ²			0.533	0.533	0.216	0.216
Controls		X		X		X
Group FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table presents estimates of Equation 1 for the Cadbury Schweppes setting. Panel A reports results for the count of CFC subsidiaries, and Panel B reports results for the share of CFC subsidiaries, calculated as the number of CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, Columns (1) and (2) report results from a log specification, Columns (3) and (4) use the inverse hyperbolic sine (IHS) transformation, and Columns (5) and (6) use a Poisson pseudo-maximum-likelihood (PPML) specification. In Panel B, Columns (1) and (2) report a linear specification on the share variable (in levels), Columns (3) and (4) report a fractional logit specification, and Columns (5) and (6) report a PPML specification. All specifications include group and year fixed effects. Standard errors are clustered at the group level. t-statistics are reported in parentheses. The lower sample size in Panel B's logit columns reflects groups with no within-group variation in the share outcome, which are dropped from the fractional logit estimation. *** p<0.01, ** p<0.05, * p<0.1.

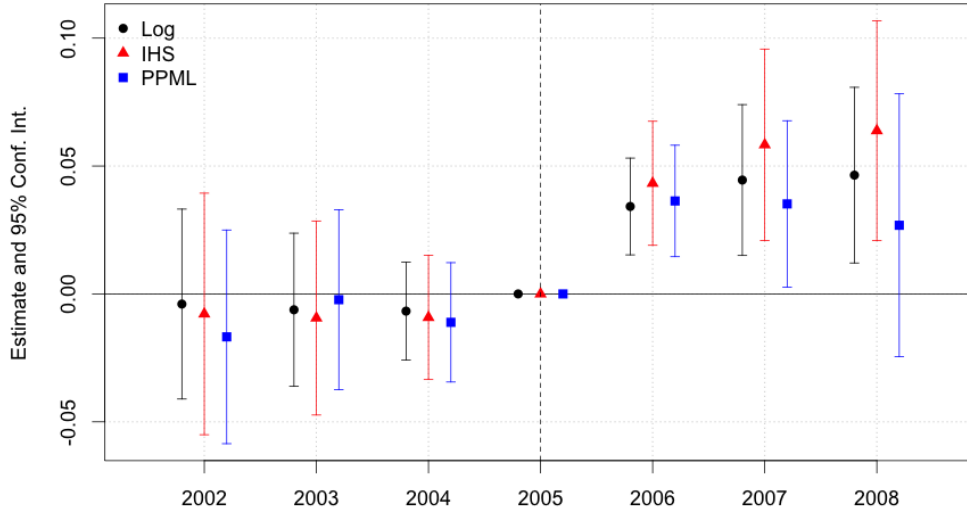
Table 5: Effects on the Count and Share of CFC Subsidiaries by Year

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Count of CFC Subsidiaries _{g,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>CS Treat_m x T-3</i>	0.0048 (0.2714)	-0.0040 (-0.2092)	0.0033 (0.1496)	-0.0078 (-0.3250)	0.0081 (0.3694)	-0.0168 (-0.7871)
<i>CS Treat_m x T-2</i>	0.0002 (0.0125)	-0.0062 (-0.4058)	-0.0016 (-0.0875)	-0.0094 (-0.4867)	0.0068 (0.3899)	-0.0022 (-0.1251)
<i>CS Treat_m x T-1</i>	-0.0004 (-0.0438)	-0.0067 (-0.6906)	-0.0012 (-0.1051)	-0.0092 (-0.7426)	0.0019 (0.1624)	-0.0111 (-0.9330)
<i>CS Treat_m x T+1</i>	0.0348*** (3.6383)	0.0342*** (3.5451)	0.0441*** (3.6006)	0.0433*** (3.4994)	0.0319*** (2.8408)	0.0363*** (3.2699)
<i>CS Treat_m x T+2</i>	0.0435*** (3.1709)	0.0445*** (2.9605)	0.0573*** (3.3043)	0.0583*** (3.0554)	0.0305* (1.8901)	0.0352** (2.1211)
<i>CS Treat_m x T+3</i>	0.0476*** (2.8803)	0.0464*** (2.6526)	0.0656*** (3.1623)	0.0638*** (2.9112)	0.0211 (0.9574)	0.0268 (1.0239)
N	8,035	8,035	8,035	8,035	8,035	8,035
Adj. R ²	0.957	0.958	0.954	0.954		
Pseudo R ²					0.718	0.718
Panel B	Share of CFC Subsidiaries _{g,t}					
	Level	Level	Logit	Logit	PPML	PPML
<i>CS Treat_m x T-3</i>	0.0065 (0.9924)	0.0058 (0.8271)	0.0577 (1.0210)	0.0620 (1.0844)	-0.0079 (-0.3626)	-0.0065 (-0.2768)
<i>CS Treat_m x T-2</i>	-0.0013 (-0.2466)	-0.0020 (-0.3572)	-0.0101 (-0.2154)	-0.0135 (-0.2888)	-0.0244 (-1.4035)	-0.0222 (-1.1719)
<i>CS Treat_m x T-1</i>	-0.0010 (-0.2831)	-0.0015 (-0.4267)	-0.0093 (-0.3171)	-0.0099 (-0.3364)	-0.0120 (-0.9841)	-0.0136 (-1.0686)
<i>CS Treat_m x T+1</i>	0.0109*** (2.9791)	0.0108*** (2.9226)	0.0919*** (2.9261)	0.0889*** (2.8533)	0.0478*** (3.9501)	0.0460*** (3.7448)
<i>CS Treat_m x T+2</i>	0.0127*** (2.6181)	0.0130** (2.4265)	0.1067*** (2.7271)	0.1042** (2.5044)	0.0646*** (4.0593)	0.0598*** (3.4435)
<i>CS Treat_m x T+3</i>	0.0127** (2.3238)	0.0137** (2.2465)	0.0995** (2.2422)	0.1065** (2.2279)	0.0732*** (4.1663)	0.0740*** (3.8445)
N	8,035	8,035	6,560	6,560	8,035	8,035
Adj. R ²	0.969	0.969				
Pseudo R ²			0.533	0.533	0.216	0.216
Controls		X		X		X
Group FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

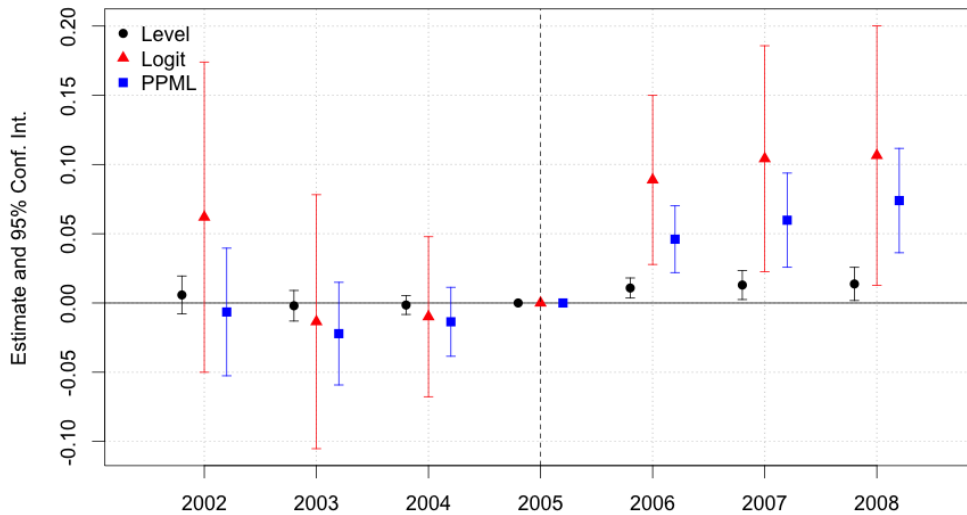
Notes: This table presents the results of estimating Equation 1 for the Cadbury Schweppes setting, where the *CS Treat_m* variable is interacted with individual year dummies instead of the *CS Post_t* indicator. Panel A reports results for the count of CFC subsidiaries, and Panel B reports results for the share of CFC subsidiaries, calculated as the number of CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, Columns (1) and (2) report results from a log specification, Columns (3) and (4) use the IHS transformation, and Columns (5) and (6) use a PPML specification. In Panel B, Columns (1) and (2) report a linear specification on the share variable (in levels), Columns (3) and (4) report a fractional logit specification, and Columns (5) and (6) report a PPML specification. All specifications include group and year fixed effects. Standard errors are clustered at the group level. t-statistics are reported in parentheses. The lower sample size in Panel B's logit columns reflects groups with no within-group variation in the share outcome, which are dropped from the fractional logit estimation. *** p<0.01, ** p<0.05, * p<0.1.

Figure 1: Event Study Effects on the Count and Share of CFC Subsidiaries

(A) Count of CFC Subsidiaries



(B) Share of CFC Subsidiaries



Notes: This figure presents event-study DiD coefficients from estimating Equation 1 for the Cadbury Schweppes setting, where the $CS\ Treat_m$ variable is interacted with individual year dummies instead of the $CS\ Post_t$ indicator. Panel A reports results for the count of CFC subsidiaries, and Panel B reports results for the share of CFC subsidiaries, calculated as the number of CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, coefficients are estimated using a log specification (black), the IHS transformation (red), and a PPML specification (blue). In Panel B, coefficients for the share variable are estimated using a linear (level) specification (black), a fractional logit specification (red), and a PPML specification (blue). All specifications include the same control variables as in Table 4, as well as group and year fixed effects. Robust standard errors clustered at the group level. The vertical lines denote 95% confidence intervals. The dashed line marks the reference year 2005.

Table 6: Cross-sectional Effects on the Count and Share of CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Count of CFC Subsidiaries _{g,t}					
	High Tax Threshold		Large MNE Size		High Sub Count	
	Yes	No	Yes	No	Yes	No
<i>CS Treat_m x CS Post_t</i>	0.0521*** (3.1989)	0.0053 (0.2318)	0.0321 (1.4621)	0.0517** (2.2885)	-0.0007 (-0.0310)	0.0911*** (4.1368)
N	6,877	3,950	3,913	3,220	3,857	3,276
Adj. R ²	0.962	0.962	0.968	0.897	0.968	0.823
Panel B	Share of CFC Subsidiaries _{g,t}					
	High Tax Threshold		Large MNE Size		High Sub Count	
	Yes	No	Yes	No	Yes	No
<i>CS Treat_m x CS Post_t</i>	0.0147** (2.5721)	-0.0008 (-0.0864)	0.0068 (1.1999)	0.0238** (2.3944)	-0.0037 (-0.7634)	0.0346*** (3.2695)
N	6,877	3,950	3,913	3,220	3,857	3,276
Adj. R ²	0.973	0.977	0.965	0.958	0.979	0.947
Controls	X	X	X	X	X	X
Group FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table reports the results for cross-sectional analyses for the Cadbury Schweppes setting based on estimates of Equation 1. Panel A reports estimates for the count of CFC subsidiaries using the log specification, while Panel B reports estimates for the share of CFC subsidiaries using a linear (level) specification. Columns (1) and (2) split treated multinationals by whether the CFC tax threshold in the shareholder country is high or low. Columns (3) and (4) split multinationals by above- versus below-median pre-treatment total assets. Columns (5) and (6) split multinationals by above- versus below-median pre-treatment number of subsidiaries. All specifications include the same control variables as in Table 4, as well as group and year fixed effects. Robust standard errors clustered at the group level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Motivational Evidence on CFC Subsidiaries' Substance

	(1)	(2)	(3)	(4)
Panel A	Log(Fixed Assets _{i,t} +1)			
<i>CS Treat_m</i>	1.5247*** (8.1517)	1.1390*** (7.0266)	1.1549*** (7.1320)	1.1295*** (6.7703)
<i>Log(Num. Empl._{i,t} + 1)</i>		1.4453*** (35.8045)	1.4542*** (36.2961)	1.4082*** (33.2373)
<i>Leverage_{i,t}</i>		0.0199*** (9.0062)	0.0200*** (9.0765)	0.0154*** (7.1722)
<i>Log(GDP_{s,t})</i>			6.5759*** (2.9627)	5.4757** (2.3244)
<i>CIT_{s,t}</i>			-0.1136*** (-2.5832)	-0.0113 (-0.2378)
N	13,108	13,108	13,108	11,701
Adj. R ²	0.202	0.304	0.304	0.308
Panel B	Log(Number of Employees _{i,t} +1)			
<i>CS Treat_m</i>	0.3175*** (5.4289)	0.0243 (0.4876)	0.0273 (0.5446)	-0.0302 (-0.5994)
<i>Log(Total Assets_{i,t})</i>		0.2203*** (32.4630)	0.2214*** (33.1100)	0.2346*** (31.5137)
<i>Leverage_{i,t}</i>		-0.0013*** (-2.8283)	-0.0013*** (-2.7531)	-0.0007 (-1.6115)
<i>Log(GDP_{s,t})</i>			-3.7184*** (-7.5854)	-0.2988 (-0.5691)
<i>CIT_{s,t}</i>			0.0513*** (5.2317)	0.0272*** (2.5916)
N	13,108	13,108	13,108	11,701
Adj. R ²	0.260	0.402	0.401	0.430
Panel C	Log(Costs of Employees _{i,t})			
<i>CS Treat_m</i>	1.1602*** (14.2104)	0.5601*** (8.4648)	0.5745*** (8.6642)	0.4394*** (6.6227)
<i>Log(Total Assets_{i,t})</i>		0.4415*** (47.3531)	0.4479*** (48.6518)	0.4778*** (46.9316)
<i>Leverage_{i,t}</i>		-0.0061*** (-7.8390)	-0.0061*** (-7.8810)	-0.0045*** (-6.2574)
<i>Log(GDP_{s,t})</i>			-1.5329* (-1.7277)	3.9142*** (4.4893)
<i>CIT_{s,t}</i>			-0.0087 (-0.5909)	0.0440*** (2.8211)
N	13,108	13,108	13,108	11,701
Adj. R ²	0.479	0.617	0.615	0.625
Host country-Year FE	X	X		
Host country-Age FE			X	
Host country FE				X
Industry-Year FE				X

Notes: This table presents the results of estimating the following regression equation for newly incorporated CFC subsidiaries in the Cadbury Schweppes setting: $Economic\ Activity_{i,t} = \beta_1 \times CS\ Treat_m + \gamma_1 \times W_{i,t} + \lambda_{s,t} + \varepsilon_{i,s,m,t}$. The economic activity outcome variables are CFC subsidiaries' fixed assets in Panel A, number of employees in Panel B, and costs of employees in Panel C. Columns (1) and (2) include host country-year fixed effects, Column (3) incorporates host country-age fixed effects, and Column (4) includes industry-year and host country fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Summary Statistics: ATAD Setting

	N	Mean	SD	P25	P50	P75
Panel A: ATAD CFC Sample						
Total Assets _{<i>i,t</i>} (in m)	37,929	19.82	38.71	0.66	3.71	15.47
Fixed Assets _{<i>i,t</i>} (in m)	37,929	7.95	18.92	0.05	0.64	4.34
Number of Employees _{<i>i,t</i>}	37,929	93.09	141.68	6.00	29.00	104.00
Costs of Employees _{<i>i,t</i>} (in m)	37,929	2.86	4.78	0.12	0.73	3.04
Leverage _{<i>i,t</i>} (in %)	34,545	5.57	18.10	0.00	0.00	0.00
GDP _{<i>i,t</i>} (in b)	37,929	430.73	750.07	81.88	129.80	341.62
CIT _{<i>i,t</i>} (in %)	37,929	16.71	5.00	10.00	19.00	20.00
Panel B: ATAD Non-CFC Sample						
Total Assets _{<i>i,t</i>} (in m)	161,753	37.92	51.35	3.34	12.01	48.29
Fixed Assets _{<i>i,t</i>} (in m)	161,753	15.67	26.45	0.30	2.25	14.86
Number of Employees _{<i>i,t</i>}	161,753	121.46	157.19	13.00	48.00	160.00
Costs of Employees _{<i>i,t</i>} (in m)	161,753	5.58	6.37	0.69	2.58	8.52
Leverage _{<i>i,t</i>} (in %)	158,958	8.68	20.46	0.00	0.00	6.36
GDP _{<i>i,t</i>} (in b)	161,753	1,487.30	941.27	442.30	1,616.37	2,227.76
CIT _{<i>i,t</i>} (in %)	161,753	29.40	5.40	25.00	30.18	33.99
Panel C: Aggregated Non-CFC Sample						
Total Assets _{<i>i,t</i>} (in m)	25,012	449.11	1,605.53	14.15	64.31	274.27
Fixed Assets _{<i>i,t</i>} (in m)	25,012	206.79	743.95	3.70	24.92	127.72
Number of Employees _{<i>i,t</i>}	25,012	900.67	3,534.09	38.00	171.00	611.00
Costs of Employees _{<i>i,t</i>} (in m)	25,012	43.90	180.27	2.16	9.28	29.51
Leverage _{<i>i,t</i>} (in %)	24,938	15.33	18.14	2.34	9.68	21.61
GDP _{<i>i,t</i>} (in b)	25,012	1,588.85	849.83	939.66	1,616.37	2,221.27
CIT _{<i>i,t</i>} (in %)	25,012	29.24	3.92	27.57	29.90	31.29

Notes: This table displays summary statistics for the ATAD setting. Panel A shows the summary statistics for the CFC subsidiary sample and Panel B for the non-CFC subsidiary sample.

Table 9: Summary Statistics: ATAD Setting, Treatment vs. Control

	N	Mean	SD	P25	P50	P75
Panel A: CFC Subsidiaries – Treatment						
Total Assets _{<i>i,t</i>} (in m)	6,211	17.13	36.62	0.56	2.79	11.38
Fixed Assets _{<i>i,t</i>} (in m)	6,211	7.54	18.51	0.05	0.63	4.17
Number of Employees _{<i>i,t</i>}	6,211	71.58	119.86	5.00	19.00	75.00
Costs of Employees _{<i>i,t</i>} (in m)	6,211	1.90	3.77	0.07	0.35	1.73
Leverage _{<i>i,t</i>} (in %)	5,576	7.61	22.01	0.00	0.00	1.51
GDP _{<i>i,t</i>} (in b)	6,211	129.00	135.45	47.27	90.85	119.01
CIT _{<i>i,t</i>} (in %)	6,211	14.15	5.58	10.00	12.50	19.00
Panel A: CFC Subsidiaries – Control						
Total Assets _{<i>i,t</i>} (in m)	31,718	20.34	39.09	0.68	3.95	16.29
Fixed Assets _{<i>i,t</i>} (in m)	31,718	8.03	19.00	0.05	0.65	4.37
Number of Employees _{<i>i,t</i>}	31,718	97.30	145.20	7.00	31.00	111.00
Costs of Employees _{<i>i,t</i>} (in m)	31,718	3.05	4.93	0.13	0.85	3.31
Leverage _{<i>i,t</i>} (in %)	28,969	5.18	17.22	0.00	0.00	0.00
GDP _{<i>i,t</i>} (in b)	31,718	489.82	804.90	82.58	160.33	456.49
CIT _{<i>i,t</i>} (in %)	31,718	17.21	4.72	16.00	19.00	20.00
Panel B: Non-CFC Subsidiaries – Treatment						
Total Assets _{<i>i,t</i>} (in m)	37,489	38.33	51.89	2.79	11.78	50.11
Fixed Assets _{<i>i,t</i>} (in m)	37,489	15.79	26.55	0.24	2.20	15.03
Number of Employees _{<i>i,t</i>}	37,489	118.00	156.68	11.00	42.00	155.00
Costs of Employees _{<i>i,t</i>} (in m)	37,489	5.40	6.32	0.63	2.32	8.31
Leverage _{<i>i,t</i>} (in %)	36,586	9.45	21.77	0.00	0.00	6.94
GDP _{<i>i,t</i>} (in b)	37,489	1,118.50	985.74	345.58	442.30	2,123.10
CIT _{<i>i,t</i>} (in %)	37,489	27.87	6.33	22.00	29.58	33.99
Panel B: Non-CFC Subsidiaries – Control						
Total Assets _{<i>i,t</i>} (in m)	124,264	37.79	51.18	3.53	12.07	47.62
Fixed Assets _{<i>i,t</i>} (in m)	124,264	15.63	26.41	0.31	2.27	14.83
Number of Employees _{<i>i,t</i>}	124,264	122.51	157.33	13.00	50.00	162.00
Costs of Employees _{<i>i,t</i>} (in m)	124,264	5.64	6.38	0.72	2.65	8.58
Leverage _{<i>i,t</i>} (in %)	122,372	8.45	20.04	0.00	0.00	6.23
GDP _{<i>i,t</i>} (in b)	124,264	1,598.56	898.19	867.10	1,674.27	2,232.82
CIT _{<i>i,t</i>} (in %)	124,264	29.86	4.99	27.81	30.18	33.99
Panel C: Aggregated Non-CFC – Treatment						
Total Assets _{<i>i,t</i>} (in m)	6,689	416.02	948.91	12.95	84.61	344.81
Fixed Assets _{<i>i,t</i>} (in m)	6,689	195.93	463.07	3.78	35.38	157.78
Number of Employees _{<i>i,t</i>}	6,689	752.29	1,735.13	25.00	149.00	696.00
Costs of Employees _{<i>i,t</i>} (in m)	6,689	37.61	84.68	1.38	8.70	36.54
Leverage _{<i>i,t</i>} (in %)	6,657	16.17	19.17	2.19	10.68	22.75
GDP _{<i>i,t</i>} (in b)	6,689	1,012.46	749.36	410.08	810.63	1,552.58
CIT _{<i>i,t</i>} (in %)	6,689	27.92	5.06	24.47	28.35	32.72
Panel C: Aggregated Non-CFC – Control						
Total Assets _{<i>i,t</i>} (in m)	18,323	461.19	1,785.95	14.53	59.14	248.85
Fixed Assets _{<i>i,t</i>} (in m)	18,323	210.75	822.91	3.67	22.39	115.46
Number of Employees _{<i>i,t</i>}	18,323	954.84	3,992.44	45.00	177.00	586.00
Costs of Employees _{<i>i,t</i>} (in m)	18,323	46.19	204.26	2.48	9.44	27.77
Leverage _{<i>i,t</i>} (in %)	18,281	15.02	17.74	2.38	9.32	21.23
GDP _{<i>i,t</i>} (in b)	18,323	1,799.26	784.43	1,287.21	1,717.52	2,437.10
CIT _{<i>i,t</i>} (in %)	18,323	29.73	3.27	28.15	30.18	31.29

Notes: This table displays summary statistics for the ATAD setting split by treatment status. Panel A shows the summary statistics for the CFC subsidiary sample and Panel B for the non-CFC subsidiary sample by treatment and control group.

Table 10: Effects on Economic Activity of CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0163 (0.2384)	-0.0001 (-0.0013)	0.0144 (0.2019)	-0.0032 (-0.0448)	0.0168 (0.5023)	0.0248 (0.7528)
<i>Log(Num. Empl._{<i>i,t-1</i>} + 1)</i>		0.4496*** (10.5286)		0.4622*** (10.3735)		0.0749*** (4.4956)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0030* (1.7064)		0.0030 (1.6184)		0.0016** (2.1031)
<i>Log(GDP_{<i>s,t</i>})</i>		0.5758 (1.2488)		0.6119 (1.2764)		-0.0466 (-0.3119)
<i>CIT_{<i>s,t</i>}</i>		0.0032 (0.6209)		0.0030 (0.5702)		0.0018 (0.7243)
N	37,929	37,929	37,929	37,929	36,839	36,839
Adj. R ²	0.893	0.895	0.888	0.890		
Pseudo R ²					0.969	0.969
Panel B	Number of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0150 (0.7814)	0.0071 (0.3951)	0.0175 (0.8306)	0.0087 (0.4398)	0.0405** (2.1721)	0.0242 (1.4735)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.1947*** (17.0586)		0.2137*** (16.7172)		0.2331*** (14.9512)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0003 (0.8506)		0.0003 (0.8324)		-0.0003 (-1.0565)
<i>Log(GDP_{<i>s,t</i>})</i>		0.1931** (2.3158)		0.2049** (2.2884)		0.1543** (1.9812)
<i>CIT_{<i>s,t</i>}</i>		-0.0012 (-0.8404)		-0.0016 (-0.9587)		-0.0004 (-0.3693)
N	37,929	37,929	37,929	37,929	37,868	37,868
Adj. R ²	0.952	0.955	0.948	0.951		
Pseudo R ²					0.947	0.950
Panel C	Costs of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0419* (1.6541)	0.0331 (1.4164)	0.0419* (1.6540)	0.0331 (1.4162)	0.1232*** (5.1462)	0.0898*** (4.2650)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.3084*** (19.8052)		0.3084*** (19.8042)		0.2925*** (19.8387)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0006 (1.2837)		0.0006 (1.2838)		-0.0002 (-0.6899)
<i>Log(GDP_{<i>s,t</i>})</i>		0.2910*** (3.0146)		0.2911*** (3.0145)		0.2385*** (2.8010)
<i>CIT_{<i>s,t</i>}</i>		0.0002 (0.1099)		0.0002 (0.1093)		-0.0004 (-0.2467)
N	37,929	37,929	37,929	37,929	37,929	37,929
Adj. R ²	0.960	0.964	0.960	0.964		
Pseudo R ²					0.981	0.983
Controls		X		X		X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table presents estimates of Equation 2 for the ATAD CFC subsidiary sample. Panel A reports results for fixed assets, Panel B for number of employees, and Panel C for costs of employees. Columns (1) and (2) report the log specification, Columns (3) and (4) the IHS transformation, and Columns (5) and (6) the PPML specification. All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. The lower sample size for the PPML specification is caused by zeroes in the respective outcome variables. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Effects on Economic Activity of CFC Subsidiaries by Year

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	-0.0633 (-0.7861)	-0.0335 (-0.4351)	-0.0666 (-0.7944)	-0.0356 (-0.4443)	0.0583 (1.4240)	0.0535 (1.3292)
<i>ATAD Treat_m x T-2</i>	0.0549 (0.7387)	0.0687 (0.9439)	0.0554 (0.7114)	0.0700 (0.9175)	0.0358 (1.1218)	0.0336 (1.0783)
<i>ATAD Treat_m x T-1</i>	-0.0426 (-0.6838)	-0.0210 (-0.3408)	-0.0463 (-0.7038)	-0.0238 (-0.3658)	0.0186 (0.6913)	0.0194 (0.7259)
<i>ATAD Treat_m x T+1</i>	0.0023 (0.0365)	0.0020 (0.0321)	-0.0024 (-0.0364)	-0.0026 (-0.0398)	0.0035 (0.1294)	0.0062 (0.2247)
<i>ATAD Treat_m x T+2</i>	-0.0127 (-0.1589)	-0.0154 (-0.1835)	-0.0169 (-0.2010)	-0.0201 (-0.2288)	0.0220 (0.6601)	0.0297 (0.8766)
<i>ATAD Treat_m x T+3</i>	-0.0171 (-0.1966)	-0.0218 (-0.2417)	-0.0199 (-0.2179)	-0.0255 (-0.2685)	0.0514 (1.5690)	0.0607* (1.7942)
N	37,929	37,929	37,929	37,929	36,839	36,839
Adj. R ²	0.893	0.895	0.888	0.890		
Pseudo R ²					0.969	0.970
Panel B	Number of Employees _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	-0.0015 (-0.0602)	0.0153 (0.6478)	0.0006 (0.0209)	0.0187 (0.7168)	-0.0147 (-0.6334)	-0.0122 (-0.5894)
<i>ATAD Treat_m x T-2</i>	-0.0354* (-1.8319)	-0.0352* (-1.8338)	-0.0394* (-1.8380)	-0.0394* (-1.8503)	-0.0330 (-1.5917)	-0.0326* (-1.7582)
<i>ATAD Treat_m x T-1</i>	-0.0065 (-0.4241)	-0.0042 (-0.2729)	-0.0086 (-0.5037)	-0.0062 (-0.3626)	-0.0038 (-0.2877)	-0.0011 (-0.0952)
<i>ATAD Treat_m x T+1</i>	0.0182 (1.2376)	0.0226 (1.5217)	0.0216 (1.3149)	0.0264 (1.5911)	0.0328** (2.1843)	0.0313** (2.1031)
<i>ATAD Treat_m x T+2</i>	0.0081 (0.4221)	0.0059 (0.3080)	0.0111 (0.5134)	0.0080 (0.3765)	0.0231 (1.1856)	0.0059 (0.3138)
<i>ATAD Treat_m x T+3</i>	-0.0005 (-0.0223)	-0.0116 (-0.5561)	-0.0004 (-0.0155)	-0.0131 (-0.5675)	0.0281 (1.4639)	0.0062 (0.3683)
N	37,929	37,929	37,929	37,929	37,868	37,868
Adj. R ²	0.952	0.955	0.948	0.951		
Pseudo R ²					0.947	0.950
Panel C	Costs of Employees _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	0.0072 (0.2091)	0.0358 (1.1354)	0.0072 (0.2095)	0.0358 (1.1356)	-0.0102 (-0.3862)	-0.0057 (-0.2407)
<i>ATAD Treat_m x T-2</i>	0.0400 (1.4351)	0.0413 (1.5370)	0.0400 (1.4351)	0.0413 (1.5370)	-0.0080 (-0.3423)	-0.0009 (-0.0439)
<i>ATAD Treat_m x T-1</i>	0.0125 (0.5499)	0.0162 (0.7263)	0.0125 (0.5498)	0.0162 (0.7261)	-0.0011 (-0.0603)	0.0082 (0.4711)
<i>ATAD Treat_m x T+1</i>	0.0427* (1.8623)	0.0495** (2.1569)	0.0427* (1.8625)	0.0495** (2.1570)	0.0915*** (5.8065)	0.0825*** (5.5206)
<i>ATAD Treat_m x T+2</i>	0.0527* (1.6905)	0.0568* (1.8742)	0.0527* (1.6899)	0.0568* (1.8734)	0.1180*** (5.8594)	0.0887*** (4.8858)
<i>ATAD Treat_m x T+3</i>	0.0775** (2.4781)	0.0673** (2.2465)	0.0775** (2.4776)	0.0673** (2.2459)	0.1437*** (6.4861)	0.1115*** (5.3977)
N	37,929	37,929	37,929	37,929	37,929	37,929
Adj. R ²	0.960	0.964	0.960	0.964		
Pseudo R ²					0.981	0.983
Controls		X		X		X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table presents the results of estimating the same specifications as Table 10, where the *ATAD Treat_m* variable is interacted with individual year dummies instead of the *ATAD Post_t* indicator. All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

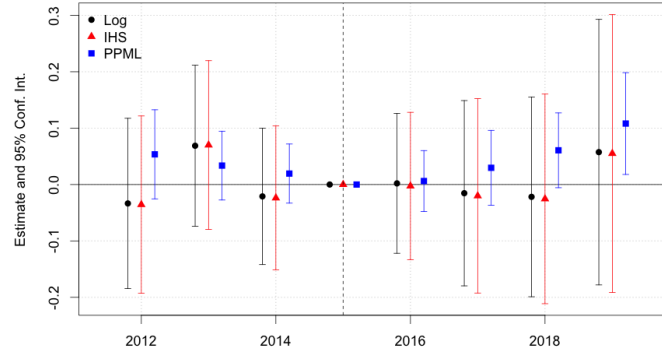
Table 12: Effects on Economic Activity of CFC Subsidiaries: Industry $\times$ Year Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0170 (0.2521)	0.0063 (0.0926)	0.0141 (0.2000)	0.0026 (0.0361)	0.0137 (0.4166)	0.0114 (0.3544)
<i>Log(Num. Empl._{<i>i,t-1</i>} + 1)</i>		0.4381*** (9.9612)		0.4502*** (9.8089)		0.0677*** (4.0149)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0049*** (3.3149)		0.0050*** (3.2378)		0.0014* (1.8060)
<i>Log(GDP_{<i>s,t</i>})</i>		0.5642 (1.3536)		0.5935 (1.3705)		0.1218 (0.7411)
<i>CIT_{<i>s,t</i>}</i>		0.0055 (1.0760)		0.0055 (1.0379)		0.0019 (0.7261)
N	37,045	37,045	37,045	37,045	36,075	36,075
Adj. R ²	0.894	0.897	0.889	0.891		
Pseudo R ²					0.970	0.970
Panel B	Number of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0192 (0.9833)	0.0090 (0.4898)	0.0218 (1.0163)	0.0105 (0.5220)	0.0420** (2.2250)	0.0249 (1.4873)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.1929*** (16.6221)		0.2113*** (16.2751)		0.2335*** (14.8607)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0003 (0.7540)		0.0003 (0.7381)		-0.0003 (-0.8759)
<i>Log(GDP_{<i>s,t</i>})</i>		0.1725* (1.9220)		0.1820* (1.8894)		0.1615** (2.0283)
<i>CIT_{<i>s,t</i>}</i>		-0.0011 (-0.6998)		-0.0013 (-0.7820)		-0.0004 (-0.3477)
N	37,045	37,045	37,045	37,045	36,993	36,993
Adj. R ²	0.953	0.956	0.949	0.952		
Pseudo R ²					0.948	0.950
Panel C	Costs of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0444* (1.7204)	0.0328 (1.3760)	0.0444* (1.7202)	0.0328 (1.3757)	0.1231*** (5.1576)	0.0879*** (4.1899)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.3063*** (19.3687)		0.3063*** (19.3676)		0.2980*** (20.3976)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0006 (1.2189)		0.0006 (1.2191)		-0.0003 (-0.9809)
<i>Log(GDP_{<i>s,t</i>})</i>		0.2436** (2.3841)		0.2436** (2.3839)		0.2703*** (3.1519)
<i>CIT_{<i>s,t</i>}</i>		0.0011 (0.5072)		0.0011 (0.5066)		0.0000 (0.0243)
N	37,045	37,045	37,045	37,045	37,045	37,045
Adj. R ²	0.961	0.964	0.961	0.964		
Pseudo R ²					0.981	0.984
Controls		X		X		X
Subsidiary FE	X	X	X	X	X	X
Industry $\times$ Year FE	X	X	X	X	X	X

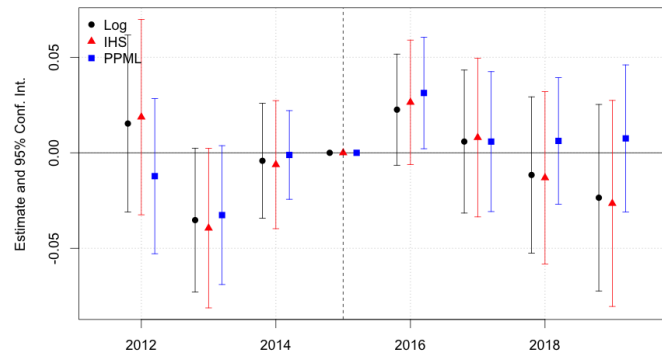
Notes: This table presents estimates of the same specifications as Table 10, replacing year fixed effects with industry-by-year fixed effects. All specifications include subsidiary fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 2: Event Study Effects on Economic Activity of CFC Subsidiaries: Year and Firm FE

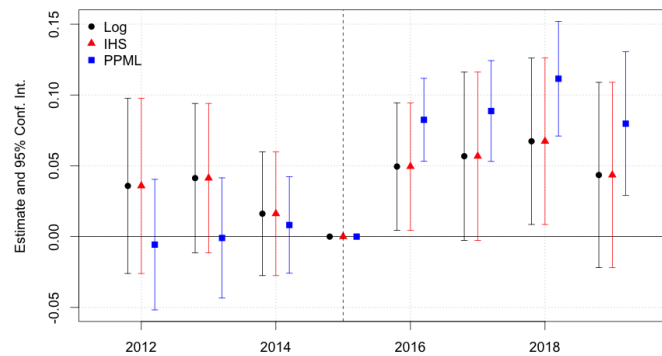
(A) Fixed Assets



(B) Number of Employees



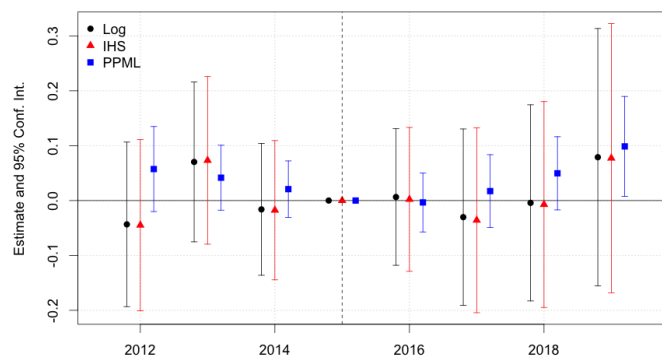
(C) Costs of Employees



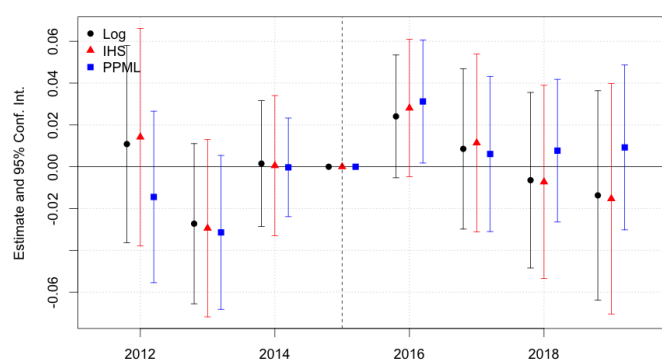
Notes: This figure presents event-study DiD coefficients from Table 11. Panels (A), (B), and (C) show fixed assets, number of employees, and costs of employees, respectively, estimated with the full set of controls using a log specification (black), the IHS transformation (red), and a PPML specification (blue). All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. Markers denote point estimates; vertical bars denote 95% confidence intervals. The dashed vertical line marks the reference year 2015.

Figure 3: Event Study Effects on Economic Activity of CFC Subsidiaries: Industry $\times$ Year and Firm FE

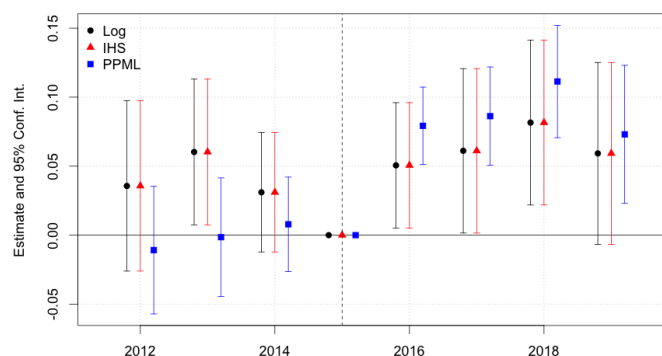
(A) Fixed Assets



(B) Number of Employees



(C) Costs of Employees



Notes: This figure presents event-study DiD coefficients analogous to Table 12, replacing year fixed effects with industry-by-year fixed effects. Panels (a), (b), and (c) show fixed assets, number of employees, and costs of employees, respectively, estimated with the full set of controls using a log specification (black), the IHS transformation (red), and a PPML specification (blue). Standard errors are clustered at the subsidiary level. Markers denote point estimates; vertical bars denote 95% confidence intervals. The dashed vertical line marks the reference year 2015.

Table 13: Effects on Economic Activity of Non-CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A						
	Fixed Assets _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0128 (0.5087)	0.0107 (0.4349)	0.0112 (0.4305)	0.0090 (0.3559)	0.0259*** (2.7526)	0.0267*** (2.8631)
<i>Log(Num. Empl._{<i>i,t-1</i>} + 1)</i>		0.4879*** (21.6977)		0.4958*** (21.3471)		0.0830*** (12.1186)
<i>Leverage_{<i>i,t-1</i>}</i>		0.0020*** (4.1473)		0.0019*** (3.9848)		0.0014*** (8.7745)
<i>Log(GDP_{<i>s,t</i>})</i>		-0.3202 (-1.0894)		-0.3273 (-1.0765)		-0.0479 (-0.4548)
<i>CIT_{<i>s,t</i>}</i>		-0.0017 (-0.4895)		-0.0017 (-0.4828)		-0.0020 (-1.6033)
N	161,753	161,753	161,753	161,753	159,952	159,952
Adj. R ²	0.889	0.892	0.885	0.887		
Pseudo R ²					0.969	0.969
Panel B						
	Number of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0068 (0.8621)	-0.0033 (-0.4690)	0.0072 (0.8536)	-0.0034 (-0.4495)	0.0001 (0.0232)	-0.0075 (-1.3458)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.2888*** (40.5724)		0.3104*** (40.2989)		0.2639*** (30.0906)
<i>Leverage_{<i>i,t-1</i>}</i>		-0.0006*** (-4.3937)		-0.0007*** (-4.3089)		-0.0004*** (-4.0486)
<i>Log(GDP_{<i>s,t</i>})</i>		0.0469 (0.4963)		0.0316 (0.3080)		-0.0300 (-0.4083)
<i>CIT_{<i>s,t</i>}</i>		-0.0002 (-0.1955)		-0.0003 (-0.2994)		0.0001 (0.1419)
N	161,753	161,753	161,753	161,753	161,686	161,686
Adj. R ²	0.960	0.964	0.958	0.962		
Pseudo R ²					0.944	0.947
Panel C						
	Costs of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0208** (2.3081)	0.0063 (0.7864)	0.0208** (2.3084)	0.0063 (0.7868)	0.0029 (0.5202)	-0.0040 (-0.8205)
<i>Log(Total Assets_{<i>i,t-1</i>})</i>		0.3580*** (40.3980)		0.3580*** (40.3963)		0.2628*** (35.3974)
<i>Leverage_{<i>i,t-1</i>}</i>		-0.0005*** (-2.9365)		-0.0005*** (-2.9358)		-0.0005*** (-4.7384)
<i>Log(GDP_{<i>s,t</i>})</i>		0.3722*** (3.4096)		0.3722*** (3.4095)		0.2349*** (3.2091)
<i>CIT_{<i>s,t</i>}</i>		-0.0009 (-0.7143)		-0.0009 (-0.7143)		0.0009 (1.2397)
N	161,753	161,753	161,753	161,753	161,753	161,753
Adj. R ²	0.948	0.952	0.948	0.952		
Pseudo R ²					0.974	0.978
Controls		X		X		X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table presents estimates of Equation 2 for the ATAD Non-CFC (placebo) sample. Panel A reports results for fixed assets, Panel B for number of employees, and Panel C for costs of employees. Columns (1) and (2) report the log specification, Columns (3) and (4) the IHS transformation, and Columns (5) and (6) the PPML specification. All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

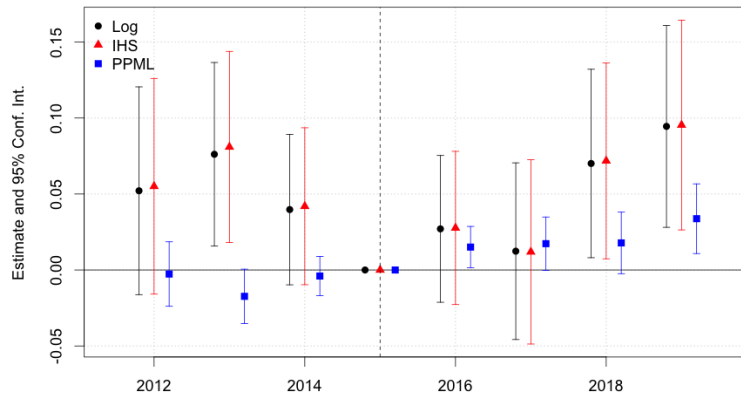
Table 14: Effects on Economic Activity of Non-CFC Subsidiaries by Year

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	0.0446 (1.2738)	0.0521 (1.4951)	0.0474 (1.3058)	0.0551 (1.5214)	-0.0052 (-0.4816)	-0.0027 (-0.2473)
<i>ATAD Treat_m x T-2</i>	0.0716** (2.3045)	0.0761** (2.4698)	0.0763** (2.3613)	0.0809** (2.5226)	-0.0200** (-2.2023)	-0.0173* (-1.9008)
<i>ATAD Treat_m x T-1</i>	0.0394 (1.5497)	0.0398 (1.5739)	0.0415 (1.5689)	0.0419 (1.5920)	-0.0046 (-0.7087)	-0.0040 (-0.6050)
<i>ATAD Treat_m x T+1</i>	0.0255 (1.0461)	0.0271 (1.0982)	0.0261 (1.0253)	0.0277 (1.0757)	0.0120* (1.7682)	0.0151** (2.1741)
<i>ATAD Treat_m x T+2</i>	0.0118 (0.4008)	0.0124 (0.4172)	0.0114 (0.3694)	0.0119 (0.3851)	0.0148* (1.6487)	0.0173* (1.9345)
<i>ATAD Treat_m x T+3</i>	0.0705** (2.2071)	0.0701** (2.2155)	0.0723** (2.1773)	0.0718** (2.1839)	0.0170 (1.6312)	0.0178* (1.7257)
N	161,753	161,753	161,753	161,753	159,952	159,952
Adj. R ²	0.889	0.892	0.885	0.887		
Pseudo R ²					0.969	0.969
Panel B	Number of Employees _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	-0.0157* (-1.6485)	-0.0062 (-0.6899)	-0.0176* (-1.6914)	-0.0076 (-0.7740)	-0.0088 (-1.2572)	-0.0003 (-0.0463)
<i>ATAD Treat_m x T-2</i>	-0.0008 (-0.1116)	0.0049 (0.6707)	-0.0000 (-0.0059)	0.0060 (0.7523)	-0.0083 (-1.4125)	-0.0027 (-0.4735)
<i>ATAD Treat_m x T-1</i>	0.0004 (0.0667)	0.0018 (0.3030)	0.0009 (0.1436)	0.0024 (0.3677)	-0.0038 (-0.8917)	0.0019 (0.4421)
<i>ATAD Treat_m x T+1</i>	-0.0073 (-1.1465)	-0.0094 (-1.4923)	-0.0084 (-1.1972)	-0.0106 (-1.5243)	-0.0051 (-1.1357)	-0.0064 (-1.4046)
<i>ATAD Treat_m x T+2</i>	0.0013 (0.1722)	-0.0047 (-0.6553)	0.0015 (0.1843)	-0.0048 (-0.6137)	-0.0033 (-0.6207)	-0.0063 (-1.2011)
<i>ATAD Treat_m x T+3</i>	0.0034 (0.4023)	-0.0035 (-0.4443)	0.0043 (0.4677)	-0.0031 (-0.3603)	-0.0093 (-1.4325)	-0.0129** (-2.0590)
N	161,753	161,753	161,753	161,753	161,686	161,686
Adj. R ²	0.960	0.964	0.958	0.962		
Pseudo R ²					0.944	0.947
Panel C	Costs of Employees _{i,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x T-3</i>	-0.0280** (-2.5257)	-0.0122 (-1.1562)	-0.0280** (-2.5258)	-0.0122 (-1.1564)	-0.0184*** (-2.7896)	-0.0077 (-1.2235)
<i>ATAD Treat_m x T-2</i>	-0.0061 (-0.6341)	0.0031 (0.3415)	-0.0061 (-0.6343)	0.0031 (0.3413)	-0.0162*** (-3.0056)	-0.0086* (-1.6451)
<i>ATAD Treat_m x T-1</i>	0.0049 (0.6593)	0.0074 (1.0152)	0.0049 (0.6591)	0.0074 (1.0149)	-0.0018 (-0.4985)	0.0046 (1.2478)
<i>ATAD Treat_m x T+1</i>	0.0015 (0.1889)	-0.0004 (-0.0539)	0.0015 (0.1892)	-0.0004 (-0.0535)	-0.0081** (-2.3070)	-0.0081** (-2.1890)
<i>ATAD Treat_m x T+2</i>	0.0118 (1.2720)	0.0049 (0.5562)	0.0118 (1.2724)	0.0050 (0.5567)	-0.0092** (-1.9882)	-0.0106** (-2.2799)
<i>ATAD Treat_m x T+3</i>	0.0186* (1.7673)	0.0090 (0.9107)	0.0186* (1.7672)	0.0090 (0.9106)	-0.0056 (-0.9965)	-0.0064 (-1.2000)
N	161,753	161,753	161,753	161,753	161,753	161,753
Adj. R ²	0.948	0.952	0.948	0.952		
Pseudo R ²					0.974	0.978
Controls		X		X		X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

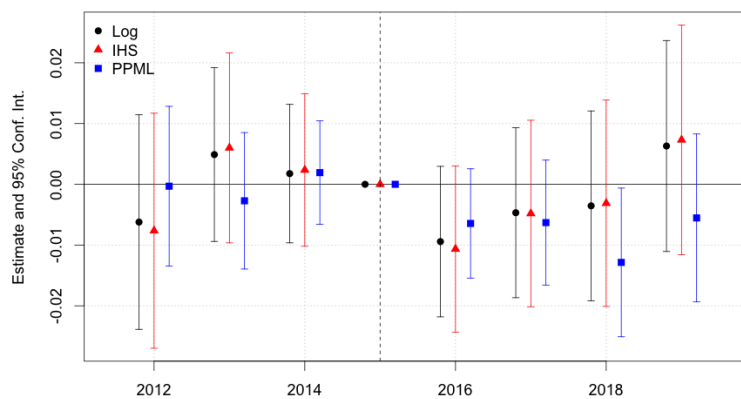
Notes: This table presents the results of estimating the same specifications as Table 13, where the *ATAD Treat_m* variable is interacted with individual year dummies instead of the *ATAD Post_t* indicator. All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 4: Event Study Effects on Economic Activity of Non-CFC Subsidiaries: Year and Firm FE

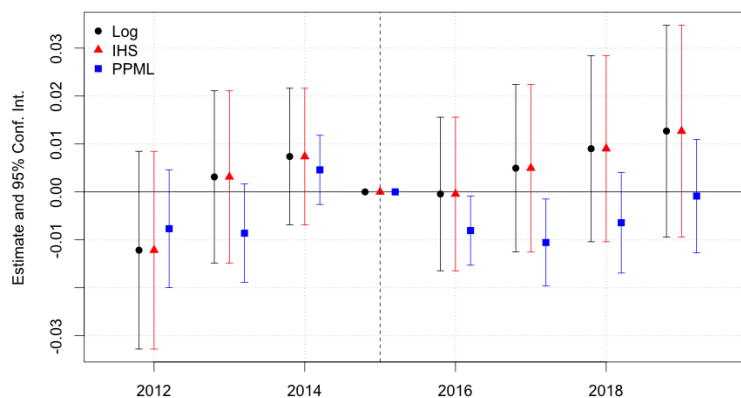
(A) Fixed Assets



(B) Number of Employees



(C) Costs of Employees



Notes: This figure presents event-study DiD coefficients from Table 14. Panels (a), (b), and (c) show fixed assets, number of employees, and costs of employees, respectively, estimated with the full set of controls using a log specification (black), the IHS transformation (red), and a PPML specification (blue). All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. Markers denote point estimates; vertical bars denote 95% confidence intervals. The dashed vertical line marks the reference year 2015.

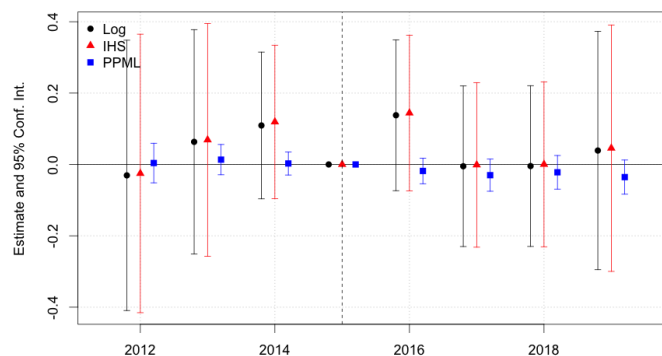
Table 15: Effects on Aggregated Economic Activity of Non-CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	0.0066 (0.0605)	0.0095 (0.0917)	0.0057 (0.0509)	0.0099 (0.0931)	-0.0360 (-1.4538)	-0.0306 (-1.2699)
<i>Log(Group Num. Empl._{<i>t-1</i>} + 1)</i>		0.1139*** (3.0468)		0.1138*** (2.9844)		0.0348*** (2.6821)
<i>Group Leverage_{<i>t-1</i>}</i>		0.0003 (0.0912)		0.0003 (0.0942)		-0.0006 (-0.8519)
<i>Log(GDP_{<i>s,t</i>})</i>		-0.2063 (-0.2002)		-0.2364 (-0.2215)		-0.3096** (-2.0315)
<i>CIT_{<i>s,t</i>}</i>		0.0104 (0.8048)		0.0104 (0.7967)		0.0088*** (2.7222)
N	7,312	7,312	7,312	7,312	7,297	7,297
Adj. R ²	0.914	0.915	0.911	0.911		
Pseudo R ²					0.991	0.992
Panel B	Number of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	-0.0888* (-1.9619)	-0.0908** (-1.9842)	-0.0931* (-1.9511)	-0.0949** (-1.9789)	0.0018 (0.0514)	-0.0054 (-0.1569)
<i>Log(Group Total Assets_{<i>t-1</i>})</i>		0.0984*** (2.7449)		0.1030*** (2.7593)		0.0353* (1.7472)
<i>Group Leverage_{<i>t-1</i>}</i>		-0.0007 (-0.5733)		-0.0006 (-0.5116)		-0.0008 (-1.0780)
<i>Log(GDP_{<i>s,t</i>})</i>		-0.4455 (-1.5615)		-0.4757 (-1.5235)		-0.5144*** (-3.3323)
<i>CIT_{<i>s,t</i>}</i>		0.0375*** (5.2913)		0.0392*** (5.2839)		0.0290*** (6.6312)
N	7,224	7,224	7,224	7,224	7,222	7,222
Adj. R ²	0.942	0.943	0.939	0.939		
Pseudo R ²					0.987	0.988
Panel C	Costs of Employees _{<i>i,t</i>}					
	Log	Log	IHS	IHS	PPML	PPML
<i>ATAD Treat_m x ATAD Post_t</i>	-0.0880** (-2.3858)	-0.0983*** (-2.6418)	-0.0880** (-2.3860)	-0.0983*** (-2.6419)	-0.0098 (-0.3935)	-0.0126 (-0.6722)
<i>Log(Group Total Assets_{<i>t-1</i>})</i>		0.1170*** (4.0109)		0.1170*** (4.0107)		0.0459** (2.4845)
<i>Group Leverage_{<i>t-1</i>}</i>		-0.0002 (-0.1805)		-0.0002 (-0.1806)		-0.0001 (-0.0851)
<i>Log(GDP_{<i>s,t</i>})</i>		-0.0332 (-0.1310)		-0.0332 (-0.1311)		-0.2206 (-1.5691)
<i>CIT_{<i>s,t</i>}</i>		0.0086 (1.2736)		0.0086 (1.2735)		0.0082*** (3.0790)
N	7,070	7,070	7,070	7,070	7,070	7,070
Adj. R ²	0.950	0.950	0.950	0.950		
Pseudo R ²					0.995	0.995
Controls		X		X		X
Parent FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

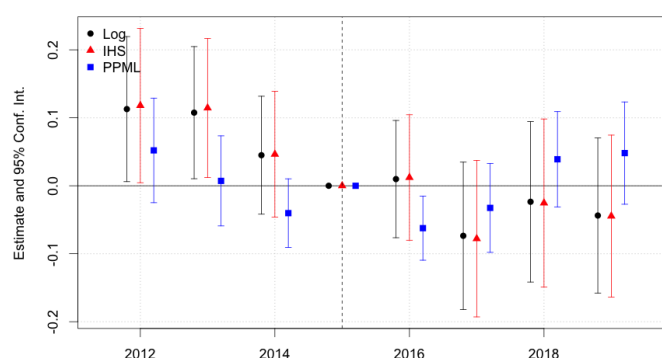
Notes: This table presents estimates of Equation 2 for a parent-year version of the Non-CFC sample, where each outcome is summed across a parent multinational's Non-CFC subsidiaries by parent-year. The treatment indicator (*ATAD Treat_m*) is a parent-level flag for whether the parent's CFC subsidiary is itself treated. Panel A reports results for fixed assets, Panel B for number of employees, and Panel C for costs of employees. Columns (1) and (2) report the log specification, Columns (3) and (4) the IHS transformation, and Columns (5) and (6) the PPML specification. Odd-numbered columns include no controls; even-numbered columns add the full set of group-consolidated controls. All specifications include parent and year fixed effects. Standard errors are clustered at the parent level; t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 5: Event Study Effects on Aggregated Economic Activity of Non-CFC Subsidiaries

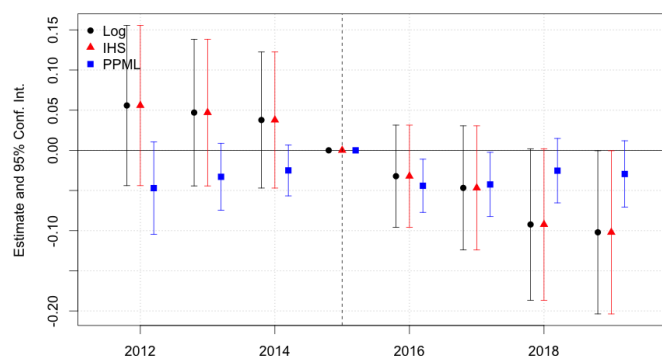
(A) Fixed Assets



(B) Number of Employees



(C) Costs of Employees



Notes: This figure presents event-study DiD coefficients from Table 15. Panels (a), (b), and (c) show fixed assets, number of employees, and costs of employees, respectively, each summed across a parent's Non-CFC subsidiaries by parent-year and estimated with the full set of group-consolidated controls using a log specification (black), the IHS transformation (red), and a PPML specification (blue). Standard errors are clustered at the parent level. Markers denote point estimates; vertical bars denote 95% confidence intervals. The dashed vertical line marks the reference year 2015.

Table 16: Heterogeneity in Effects on CFC Substance

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	-0.0229 (-0.3005)	0.1239 (1.4909)	0.0194 (0.2321)	0.0183 (0.1693)	0.1424 (1.1386)	-0.0594 (-1.0934)
N	27,034	30,529	21,826	22,719	25,224	32,017
Adj. R ²	0.894	0.897	0.905	0.883	0.807	0.769
Panel B	Number of Employees _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	0.0239 (1.2526)	0.0951*** (3.4952)	0.0158 (0.6474)	0.0322 (1.2405)	0.0847*** (3.8337)	-0.0146 (-0.6889)
N	27,227	31,673	21,700	23,099	25,042	28,121
Adj. R ²	0.955	0.969	0.957	0.964	0.865	0.899
Panel C	Costs of Employees _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	0.0193 (0.6136)	0.0386 (0.9144)	0.0465 (1.1238)	0.0529 (1.6228)	0.0940** (2.5202)	-0.0363 (-1.0640)
N	23,801	30,087	23,696	24,079	24,742	27,293
Adj. R ²	0.951	0.950	0.949	0.959	0.883	0.893
Controls	X	X	X	X	X	X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table reports the results of heterogeneity analyses based on estimates of Equation 2 for subsamples of CFC subsidiaries, using a log specification with the full set of controls. Panel A reports estimates for fixed assets, Panel B for number of employees, and Panel C for costs of employees. Columns (1) and (2) split the sample by whether the tax rate differential between the host country and the multinational's home country is above (Yes) or below (No) the median. Columns (3) and (4) present estimates for subsidiaries with above- (Yes) and below-median (No) shares of passive income, measured as the pre-treatment average ratio of financial revenue to total revenue. Columns (5) and (6) differentiate subsidiaries based on pre-existing levels of economic activities. All specifications include subsidiary and year fixed effects. Standard errors are clustered at the subsidiary level. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 17: Heterogeneity in Effects on CFC Substance: PPML

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Fixed Assets _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	0.0081 (0.2360)	-0.0476* (-1.8078)	-0.0067 (-0.2117)	-0.0177 (-0.4026)	-0.1826 (-1.0321)	-0.0147 (-0.6644)
N	25,501	29,412	21,045	21,828	22,651	31,965
Pseudo R ²	0.971	0.972	0.970	0.969	0.922	0.961
Panel B	Number of Employees _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	0.0388** (2.2276)	0.0094 (0.3094)	0.0339* (1.8003)	0.0201 (0.7824)	0.0542 (0.7994)	0.0219 (1.4590)
N	25,470	30,384	20,424	22,357	22,097	28,112
Pseudo R ²	0.951	0.953	0.952	0.950	0.698	0.929
Panel C	Costs of Employees _{<i>i,t</i>}					
	High Tax Diff. Yes	No	High Passive Income Yes	No	Low Substance Yes	No
<i>ATAD Treat_m x ATAD Post_t</i>	0.0460* (1.9445)	0.0551 (1.5926)	0.0851*** (3.1265)	0.0274 (0.9436)	0.0680 (1.5675)	0.0599*** (2.9374)
N	23,801	30,087	23,696	24,079	24,742	27,293
Pseudo R ²	0.983	0.978	0.979	0.983	0.919	0.965
Controls	X	X	X	X	X	X
Subsidiary FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

Notes: This table reports estimates of the same specifications as Table 16, using a Poisson pseudo-maximum-likelihood (PPML) specification instead of the log specification. Standard errors are clustered at the subsidiary level; t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Appendix

Table A-1: Variable Definitions

Variable	Definition
Panel A: Cadbury Schweppes Sample	
CS Treat _m	Indicator variable equal to one if the multinational is headquartered in a country m that has CFC rules implemented at the time of the Cadbury Schweppes ruling, and zero otherwise.
CS Post _t	Indicator variable set equal to one if year t is after 2005, and zero otherwise.
Number Total Incorp. _{g,t}	Multinational group g's number of total incorporations worldwide in year t.
Number CFC Incorp. _{g,t}	Multinational group g's number of CFC incorporations in the EEA in year t.
Number Non-CFC Incorp. _{g,t}	Multinational group g's number of non-CFC incorporations in the EEA in year t.
Share CFC Incorp. _{g,t}	Share of CFC incorporations in the EEA relative to total incorporations worldwide within multinational group g in year t.
Share Non-CFC Incorp. _{g,t}	Share of non-CFC incorporations in the EEA relative to total incorporations worldwide within multinational group g in year t.
Total Assets _{g,t}	Multinational group g's consolidated total assets in year t.
Number of Employees _{g,t}	Multinational group g's consolidated number of employees in year t.
Leverage _{g,t}	Multinational group g's leverage in year t, calculated as consolidated long term debt divided by consolidated total assets times 100.
GDP _{m,t}	Total GDP in home country m in year t.
CIT _{m,t}	CIT rate applicable in home country m in year t in percent.
Panel B: ATAD Sample	
ATAD Treat _m	Indicator equal to one if the subsidiary's multinational is headquartered in a country m that newly implemented CFC rules due to the ATAD, and zero otherwise.
ATAD Post _t	Indicator variable set equal to one if year t is after 2015, and zero otherwise.
Total Assets _{i,t}	Subsidiary i's unconsolidated total assets in year t.
Fixed Assets _{i,t}	Subsidiary i's unconsolidated fixed assets in year t.
Number of Employees _{i,t}	Subsidiary i's unconsolidated number of employees in year t.
Costs of Employees _{i,t}	Subsidiary i's unconsolidated costs of employees in year t.
Leverage _{i,t}	Subsidiary i's leverage in year t, calculated as unconsolidated long term debt divided by unconsolidated total assets times 100.
GDP _{s,t}	Total GDP in host country s in year t.
CIT _{s,t}	CIT rate applicable in host country s in year t in percent.

Table A-2: CFC Rules in the EU

EU Countries	Enactment of CFC Rules	Cadbury Schweppes Setting			ATAD Setting		
		Treatment Group	Control Group	Tax Threshold (in %)	Treatment Group	Control Group	Tax Threshold (in %)
Austria (AT)	2019		X		X		12.50
Belgium (BE)	2019		X		X		14.79
Bulgaria (BG)	2019		X		X		5.00
Croatia (HR)	2019		X		X		9.00
Cyprus (CY)	2019		X		X		6.25
Czech Republic (CZ)	2019		X		X		9.50
Denmark (DK)	1995	X		21.00	-	-	-
Estonia (EE)	2019		X		-	-	-
Finland (FI)	1995	X		15.60		X	12.00
France (FR)	1980	X		17.48		X	17.22
Germany (DE)	1972	X		25.00		X	25.00
Greece (GR)	2013		X		-	-	-
Hungary (HU)	1996	X		10.67		X	4.50
Ireland (IE)	2019		X		X		6.25
Italy (IT)	2002	X				X	13.90
Latvia (LV)	2019		X		-	-	-
Lithuania (LT)	2002	X		11.25		X	7.50
Luxembourg (LU)	2019		X		X		12.47
Malta (MT)	2019		X		X		17.50
Netherlands (NL)	2019		X		X		9.00
Poland (PL)	2015		X		-	-	-
Portugal (PT)	1995	X		15.00		X	15.75
Romania (RO)	2018		X		X		8.00
Slovak Republic (SK)	2019		X		X		10.50
Slovenia (SI)	2019		X		X		9.50
Spain (ES)	1994	X		26.25		X	18.75
Sweden (SE)	1991	X		15.40		X	11.77
United Kingdom (GB)	1984	X		22.50	-	-	-

Notes: This table summarizes the CFC regimes in force at the time of the Cadbury Schweppes ruling in 2006 and at the entry into force of the ATAD CFC provisions in 2019. Countries shown in bold additionally rely on black or white lists to identify low-tax jurisdictions. The tax threshold indicates the CIT rate below which subsidiaries are classified as low-tax. Italy did not specify an explicit tax threshold in 2006 and instead relied exclusively on a black list. Countries marked with a dash are excluded from the ATAD analysis because they either introduced CFC rules during the pre-treatment period (Greece and Poland) or did not specify a tax threshold (Denmark, Estonia, and Latvia). Great Britain is not considered as EU country in the ATAD setting due to the Brexit.

Table A-3: Summary Statistics Cadbury Schweppes - Treatment vs. Control

	N	Mean	SD	P25	P50	P75
Panel A: Treated Multinationals (CS Treat = 1)						
Number Total Subs. _{<i>g,t</i>}	5,243	35.76	70.71	5.00	12.00	33.00
Number Total EEA Subs. _{<i>g,t</i>}	5,243	32.84	66.94	4.00	11.00	31.00
Number CFC Subs. _{<i>g,t</i>}	5,243	3.93	6.70	1.00	2.00	3.00
Number Non-CFC Subs. _{<i>g,t</i>}	5,243	28.91	62.41	3.00	9.00	26.00
Total Assets _{<i>g,t-1</i>} (in m)	5,243	3,359.70	6,179.86	97.55	448.02	2,670.51
Number of Employees _{<i>g,t-1</i>}	5,243	11,879.62	20,225.06	465.50	2,136.00	11,436.00
Leverage _{<i>g,t-1</i>} (in %)	5,243	16.66	15.43	4.47	13.29	24.39
GDP _{<i>m,t</i>} (in b)	5,243	1,564.16	809.06	885.23	1,881.34	2,187.35
CIT _{<i>m,t</i>} (in %)	5,243	32.22	4.29	30.00	30.00	35.43
Panel B: Control Multinationals (CS Treat = 0)						
Number Total Subs. _{<i>g,t</i>}	2,792	16.00	32.57	2.00	5.00	15.00
Number Total EEA Subs. _{<i>g,t</i>}	2,792	14.86	30.38	2.00	5.00	13.00
Number CFC Subs. _{<i>g,t</i>}	2,792	5.36	10.79	1.00	2.00	5.00
Number Non-CFC Subs. _{<i>g,t</i>}	2,792	9.50	22.99	0.00	1.00	8.00
Total Assets _{<i>g,t-1</i>} (in m)	2,792	1,220.29	3,209.32	39.73	128.12	687.34
Number of Employees _{<i>g,t-1</i>}	2,792	4,922.35	12,450.36	203.75	687.50	2,837.00
Leverage _{<i>g,t-1</i>} (in %)	2,792	15.17	15.55	2.66	11.20	22.84
GDP _{<i>m,t</i>} (in b)	2,792	334.76	174.93	194.07	292.38	539.55
CIT _{<i>m,t</i>} (in %)	2,792	26.82	8.00	19.00	28.00	33.99

Notes: The table displays the summary statistics for the multinational sample considered in the Cadbury Schweppes setting. Panel A shows summary statistics for treated multinationals and Panel B shows summary statistics for control multinationals. Total assets and fixed assets are denoted in million EUR. GDP is expressed in billion EUR.

Table A-4: Effects on the Count and Share of CFC Subsidiaries - Industry-Year Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Count of CFC Subsidiaries _{g,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	0.0478*** (3.1027)	0.0510*** (3.1909)	0.0642*** (3.3307)	0.0680*** (3.3819)	0.0488** (2.2769)	0.0500** (2.2523)
N	7,947	7,947	7,947	7,947	7,944	7,944
Adj. R ²	0.959	0.960	0.956	0.956		
Pseudo R ²					0.720	0.721
Panel B	Share of CFC Subsidiaries _{g,t}					
	Level	Level	Logit	Logit	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	0.0125** (2.4383)	0.0141*** (2.6387)	0.1141*** (2.6329)	0.1194*** (2.7923)	0.0741*** (4.3505)	0.0722*** (4.0199)
N	7,947	7,947	6,519	6,519	7,944	7,944
Adj. R ²	0.970	0.970				
Pseudo R ²			0.534	0.534	0.216	0.216
Controls		X		X		X
Group FE	X	X	X	X	X	X
Industry x Year FE	X	X	X	X	X	X

Notes: This table presents estimates of Equation 1 for the Cadbury Schweppes setting. Panel A reports results for the count of CFC subsidiaries, and Panel B reports results for the share of CFC subsidiaries, calculated as the number of CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, Columns (1) and (2) report results from a log specification, Columns (3) and (4) use the inverse hyperbolic sine (IHS) transformation, and Columns (5) and (6) use a Poisson pseudo-maximum-likelihood (PPML) specification. In Panel B, Columns (1) and (2) report a linear specification on the share variable (in levels), Columns (3) and (4) report a fractional logit specification, and Columns (5) and (6) report a PPML specification. All specifications include group and industry-year fixed effects. Standard errors are clustered at the group level. t-statistics are reported in parentheses. The lower sample size in Panel B's logit columns reflects groups with no within-group variation in the share outcome, which are dropped from the fractional logit estimation. *** p<0.01, ** p<0.05, * p<0.1.

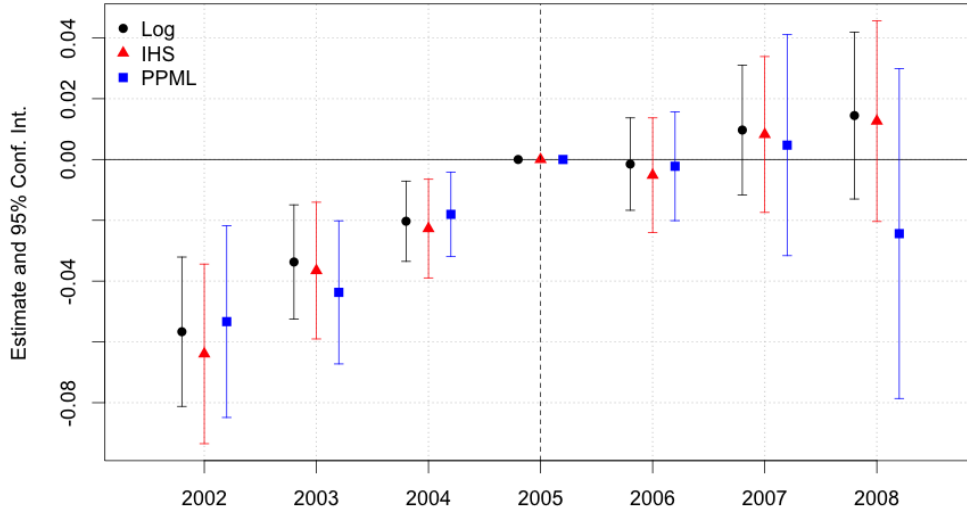
Table A-5: Effects on the Count and Share of Non-CFC Subsidiaries

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	Count of Non-CFC Subsidiaries _{g,t}					
	Log	Log	IHS	IHS	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	0.0305** (2.4966)	0.0277** (2.3685)	0.0301** (2.0486)	0.0276** (1.9790)	-0.0032 (-0.1641)	0.0161 (0.7874)
<i>Log(Total Assets_{g,t-1})</i>		0.0528*** (4.7807)		0.0639*** (4.8338)		0.0534*** (3.2774)
<i>Log(Num. Empl._{g,t-1} + 1)</i>		0.0067 (1.2342)		0.0066 (1.0571)		-0.0042 (-0.7966)
<i>Leverage_{g,t-1}</i>		-0.0000 (-0.1026)		-0.0001 (-0.2638)		0.0003 (0.8002)
<i>Log(GDP_{m,t})</i>		-0.2618*** (-3.3509)		-0.3095*** (-3.2940)		0.2290* (1.7490)
<i>CIT_{m,t}</i>		-0.0016 (-1.6133)		-0.0023* (-1.8496)		-0.0052*** (-2.7536)
N	8,035	8,035	8,035	8,035	6,560	6,560
Adj. R ²	0.994	0.994	0.994	0.994		
Pseudo R ²					0.926	0.926
Panel B	Share of Non-CFC Subsidiaries _{g,t}					
	Level	Level	Logit	Logit	PPML	PPML
<i>CS Treat_m x CS Post_t</i>	-0.0113** (-2.1611)	-0.0123** (-2.2507)	-0.0926** (-2.1250)	-0.0943** (-2.1996)	-0.0072 (-0.7164)	-0.0101 (-1.0408)
<i>Log(Total Assets_{g,t-1})</i>		-0.0089 (-1.4999)		-0.0872* (-1.6787)		-0.0136 (-1.5235)
<i>Log(Num. Empl._{g,t-1} + 1)</i>		0.0057* (1.7088)		0.0602* (1.7888)		0.0084* (1.7249)
<i>Leverage_{g,t-1}</i>		-0.0001 (-0.4530)		-0.0006 (-0.4032)		-0.0001 (-0.3485)
<i>Log(GDP_{m,t})</i>		-0.0200 (-0.5229)		-0.2371 (-0.6361)		-0.1158 (-1.4694)
<i>CIT_{m,t}</i>		0.0001 (0.2166)		0.0001 (0.0124)		0.0008 (0.7971)
N	8,035	8,035	6,560	6,560	6,560	6,560
Adj. R ²	0.969	0.969				
Pseudo R ²			0.520	0.520	0.035	0.035
Controls		X		X		X
Group FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X

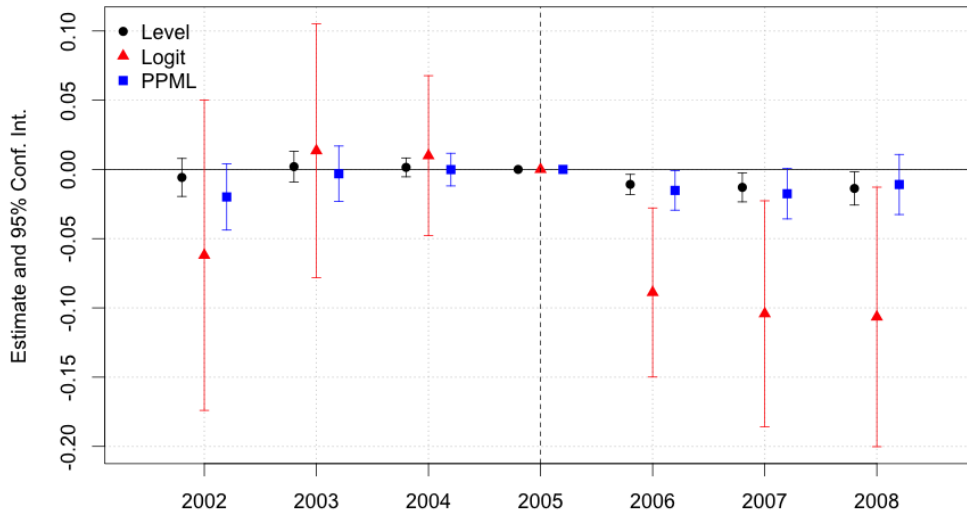
Notes: This table presents estimates of Equation 1 for the Cadbury Schweppes setting. Panel A reports results for the count of non-CFC subsidiaries, and Panel B reports results for the share of non-CFC subsidiaries, calculated as the number of non-CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, Columns (1) and (2) report results from a log specification, Columns (3) and (4) use the inverse hyperbolic sine (IHS) transformation, and Columns (5) and (6) use a Poisson pseudo-maximum-likelihood (PPML) specification. In Panel B, Columns (1) and (2) report a linear specification on the share variable (in levels), Columns (3) and (4) report a fractional logit specification, and Columns (5) and (6) report a PPML specification. All specifications include group and year fixed effects. Standard errors are clustered at the group level. t-statistics are reported in parentheses. The lower sample size in Panel A's PPML columns and Panel B's logit and PPML columns reflects groups with no within-group variation in the count or share of non-CFC subsidiaries. *** p<0.01, ** p<0.05, * p<0.1.

Figure A-1: Event Study Effects on the Count and Share of Non-CFC Subsidiaries

(A) Count of Non-CFC Subsidiaries



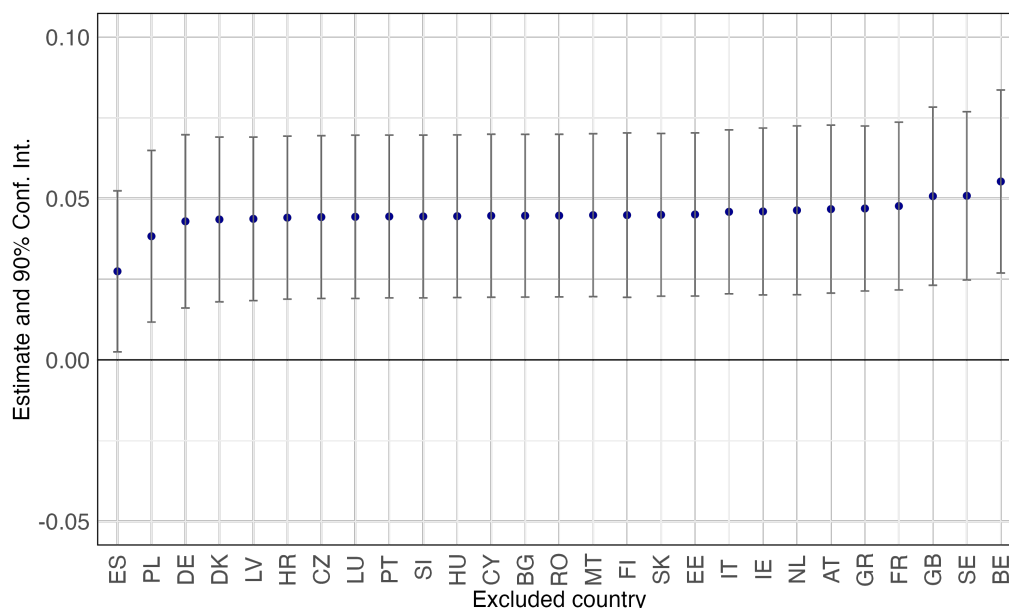
(B) Share of Non-CFC Subsidiaries



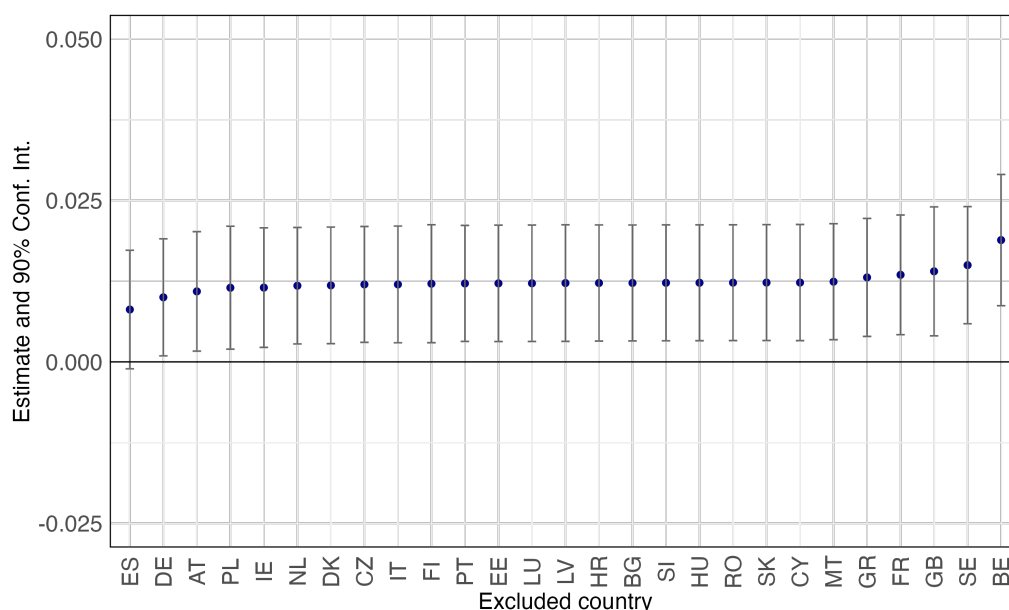
Notes: This figure presents event-study DiD coefficients from estimating Equation 1 for the Cadbury Schweppes setting, where the $CS\ Treat_m$ variable is interacted with individual year dummies instead of the $CS\ Post_t$ indicator. Panel A reports results for the count of non-CFC subsidiaries, and Panel B reports results for the share of non-CFC subsidiaries, calculated as the number of non-CFC subsidiaries divided by the total number of subsidiaries of a multinational. In Panel A, coefficients are estimated using a log specification (black), the IHS transformation (red), and a PPML specification (blue). In Panel B, coefficients for the share variable are estimated using a linear (level) specification (black), a fractional logit specification (red), and a PPML specification (blue). All specifications include the same control variables as in Table A-5, as well as group and year fixed effects. Robust standard errors clustered at the group level. The vertical lines denote 95% confidence intervals. The dashed line marks the reference year 2005.

Figure A-2: Leave-one-out Analysis Count and Share of CFC Subsidiaries

(A) Count of CFC Subsidiaries

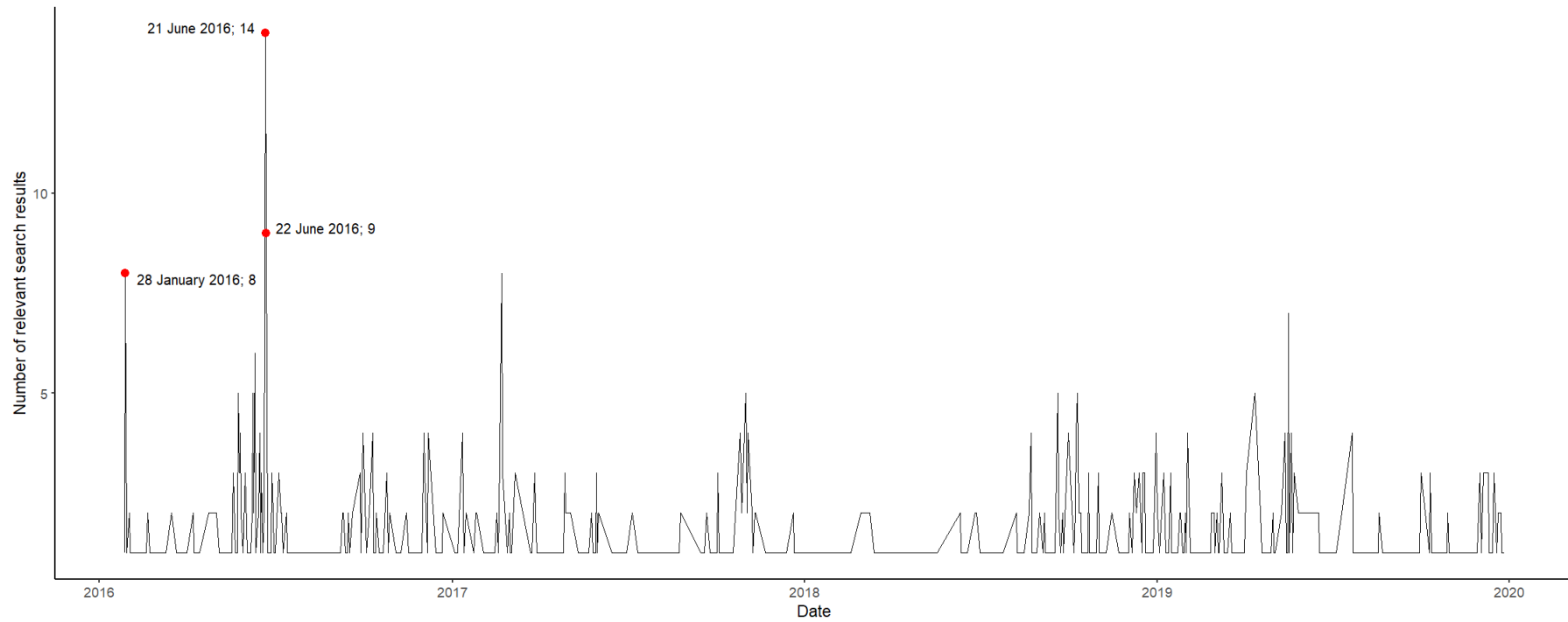


(B) Share of CFC Subsidiaries



Notes: This figure presents coefficient estimates from the leave-one-out analysis for the Cadbury Schweppes setting that sequentially excludes each home country from the sample. Panel A reports estimates for the count of CFC subsidiaries (log specification), and Panel B reports estimates for the share of CFC subsidiaries (level specification). All specifications include the same control variables as in Table 4, as well as group and year fixed effects. Standard errors are clustered at the group level. The grey vertical lines denote 90% confidence intervals.

Figure A-3: Media coverage analysis



Notes: This figure shows a media coverage analysis conducted with Factiva. We plot the number of relevant media articles over time when searching for "anti-tax avoidance directive".

Table A-6: Country-Year Distribution: ATAD CFC Sample

Subsidiary country	2012	2013	2014	2015	2016	2017	2018	2019
BG	1026	1087	1140	1238	1288	1263	1234	1213
CY	3	23	36	25	30	38	41	48
CZ	736	423	45	11	9	5	4	1
DK	31	179	205	209	209	182	187	177
EE	31	36	41	42	48	45	48	48
FI	106	141	149	164	171	171	170	170
GB	2777	2841	2891	2980	3077	3156	3209	3138
GR	79	82	107	105	105	111	112	113
HR	183	198	212	222	241	242	228	226
HU	1224	1332	1361	1363	1368	1377	1339	1220
IE	639	719	786	822	899	870	851	829
IS	11	13	12	12	13	14	14	16
LT	26	28	31	28	28	31	31	30
LU	1253	1350	1444	1786	3590	4126	4091	3950
LV	146	37	28	30	188	198	196	192
MT	339	466	595	719	832	803	733	668
NL	1376	1418	1458	1495	1577	1375	1236	1138
NO	206	221	232	238	246	245	244	244
PL	1067	1150	1194	1199	1193	1237	1458	1460
RO	250	1424	1596	1673	1734	1686	403	413
SE	243	256	279	297	314	562	562	567
SK	510	502	558	572	577	591	584	571

Notes: This table reports the number of subsidiary-year observations in the ATAD CFC sample by host country and year.